Structural Oil Shocks Across Asia

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Abstract

We estimate the effects of structurally identified oil shocks on output and inflation in eight Asian economies, using a Bayesian structural vector autoregression that separates oil-supply, global-demand, and speculative demand shocks. The effect of an oil price increase depends on its cause, and it differs across countries. A demand-driven increase raises output in six of the eight economies, because it arrives with a stronger world economy. A supply-driven increase of the same size is contractionary for some economies and close to neutral for others, and it lowers output most in Vietnam and the Philippines. The heterogeneity across countries and across shocks, rather than any single average effect, is the central result, and we use it to trace the regional consequences of concrete oil price episodes.

Keywords. Oil price shocks, structural vector autoregression, sign restrictions, block exogeneity, structural scenario analysis, Asia, emerging markets.

JEL classification. Q43, Q41, E32, F41, C32, O53.

1 Introduction

For much of the postwar period, empirical macroeconomics treated the price of oil as exogenous and read a rising oil price as unambiguously contractionary for oil importers. This reading followed the influential account of Hamilton (1983), who documented that almost every recession in the United States after the Second World War had been preceded by a sharp increase in the price of crude petroleum. A large literature then refined the size and the timing of these effects, and Kilian (2008) provides a survey. The decisive change in

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thinking came with Kilian (2009), who argued that the price of oil is an endogenous variable determined jointly with global quantities. His central message is that not all oil-price shocks are alike. An oil-price increase that is driven by a disruption to physical supply has very different macroeconomic consequences from one driven by a boom in global demand for industrial commodities.

Figure 1 makes this point with the data. It plots the real price of Brent crude since the year 2000 and marks five well-known episodes. The oil price reached comparable heights in 2008 and in 2022, yet the two episodes had opposite causes. The 2008 spike reflected booming demand from fast-growing emerging economies, while the 2022 spike reflected fears about supply after the invasion of Ukraine. In the same way, the price collapsed for very different reasons in 2014 through 2016, when new shale supply flooded the market, and in early 2020, when the pandemic destroyed demand. A single number for the oil price therefore hides a range of very different economic stories. Reducing the whole history to a correlation between the oil price and domestic activity averages over these episodes and is hard to interpret.

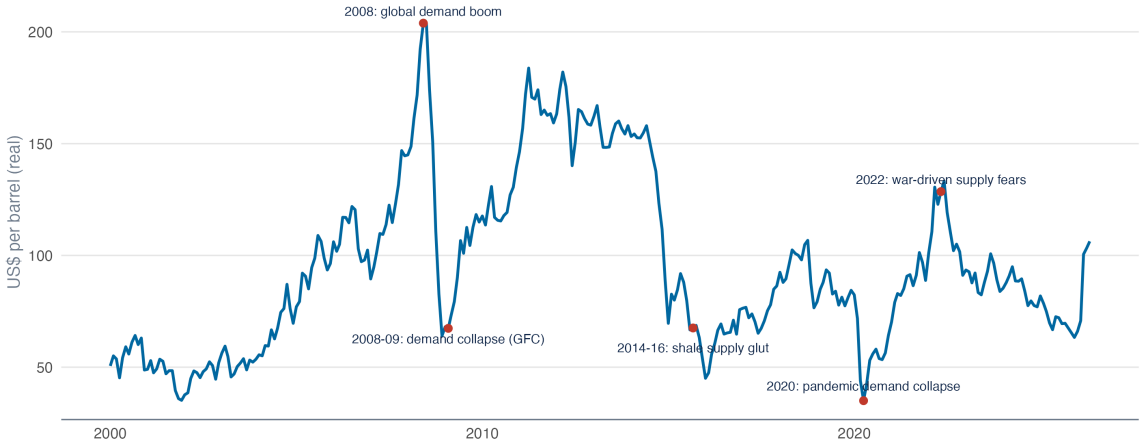


Figure 1: The real price of oil has reached similar levels for opposite reasons

Notes. The real price of Brent crude in constant United States dollars per barrel from 2000 to 2025, with five episodes marked. The causes of these episodes are discussed in the text. *Source.* Brent spot price from the Federal Reserve Economic Data service, series POILBREUSDM, deflated by the consumer price index, series CPIAUCSL.

The distinction is not merely academic. A supply-driven increase is a classic adverse terms-of-trade shock for an oil importer. It raises imported costs and inflation at the same time as it squeezes real activity, so it is stagflationary. A demand-driven increase of the same size arrives together with a stronger external environment that includes higher export demand and firmer commodity-linked incomes, so its net effect on domestic output can be neutral or even positive even as it also raises prices. A policymaker who cannot tell the two apart cannot know whether a given oil-price increase calls for a tighter stance to contain

inflation or for patience because activity is already weakening. Telling them apart requires a structural model of the global oil market that identifies the distinct shocks. Kilian and Murphy (2014) provide the workhorse. Their four-variable structural model of the global crude-oil market uses global production, global real activity, the real price of oil, and above-ground inventories to separately identify a flow-supply shock, a flow-demand shock, and a speculative demand shock. Kilian and Murphy (2012) show that sign restrictions on their own are not enough to pin down these dynamics and that bounds on the short-run price elasticities of oil supply and demand are needed to obtain sensible responses. Baumeister and Hamilton (2019) reach related conclusions with a Bayesian treatment of incomplete identification.

Figure 2 applies this apparatus to the historical record and makes the motivation quantitative. It reports a historical decomposition of the real price of oil from the four-variable structural model, in which the deviation of the real price from its mean is expressed as the sum of the contributions of the flow-supply, flow-demand, and speculative demand shocks. The figure shows that the same movement in the price has come from very different sources at different times. The run-up of 2007 and 2008 and the elevated prices of the following years were overwhelmingly a demand phenomenon, shown in blue, whereas the sharp declines of late 2008 and of early 2020 were demand collapses of the same character with the opposite sign. Flow-supply disturbances, shown in red, were a comparatively small part of the variation for most of the sample and became visible mainly around 2022. A single series for the oil price therefore superimposes several distinct structural histories, which is precisely why a structural decomposition is required before the domestic consequences can be interpreted.

Identifying the source of an oil-price movement is only the first step. The same structural oil shock can transmit very differently across economies. What matters is the structural context of each economy, which includes whether it is a net importer or a producer of oil, how energy-intensive its production is, whether retail fuel prices are administered or deregulated, the exchange-rate regime, and the credibility and the framework of monetary policy. Baumeister and Kilian (2014) show how a structural oil-market model can be used to build policy-relevant scenarios that ask what happens if the oil price follows a given path for a given reason, rather than relying on an unconditional forecast.

Emerging and advanced Asia is an especially informative setting for this question because it combines heavy exposure to imported energy with wide cross-country differences. Figure 3 shows that the Asian economies we study range from South Korea, which imports the great majority of the energy it uses, through Thailand, the Philippines, India, Vietnam, and China, and down to Malaysia and Indonesia, which are net exporters of energy once coal and gas are counted. This variation is precisely what makes a common structural framework valuable,

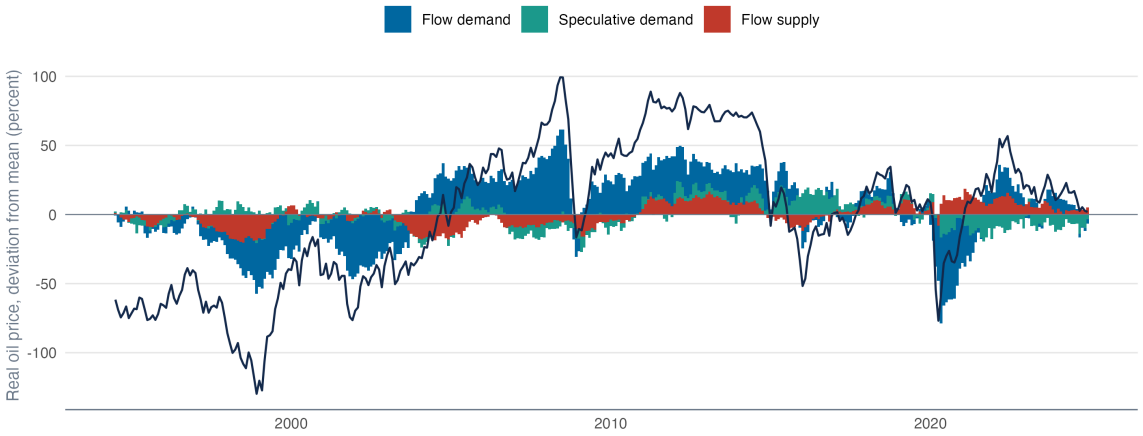


Figure 2: The same oil price has been driven by different shocks at different times

Notes. Historical decomposition of the real price of oil from the four-variable model of Kilian and Murphy (2014), estimated on monthly data from 1994 to 2024. Each bar is the posterior median contribution of a structural shock to the deviation of the real price from its mean, and the navy line is that deviation. *Source.* Authors' structural VAR on the global oil block.

because it allows the global oil block to be held identical across all the economies while the domestic transmission is permitted to differ. Any differences found across countries can then be attributed to domestic structure rather than to how the oil shock itself is measured.

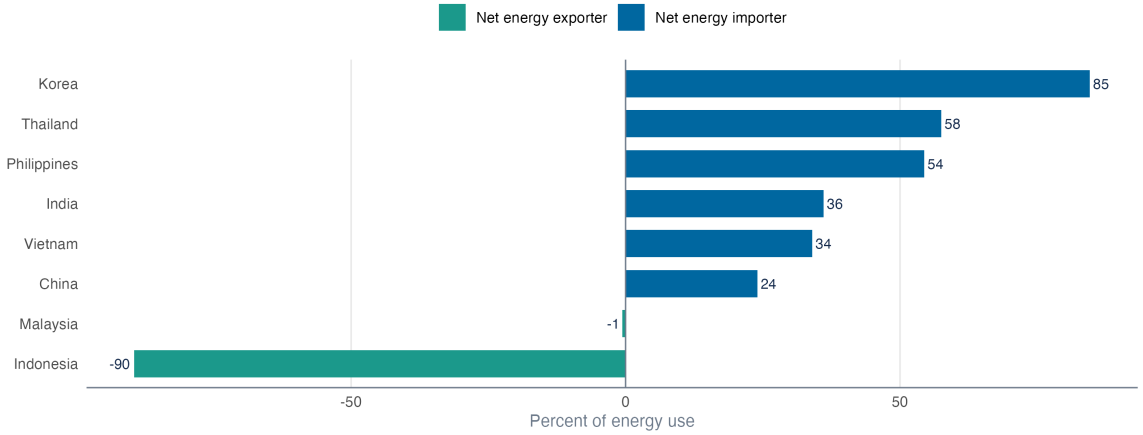


Figure 3: The economies studied are heavily but unevenly exposed to imported energy

Notes. Net energy imports as a percent of energy use, most recent year available. A positive value denotes a net importer and a negative value a net exporter. *Source.* World Bank World Development Indicators, series EG.IMP.CONS.ZS.

An earlier literature already shows that oil shocks matter for Asian economies and that their effects differ across countries, a body of work that we review in Section 2. Our departure from it is to hold the global oil block, the identification, and the definition of every domestic variable fixed across a set of eight economies, so that the differences we find can be read as

differences in domestic structure rather than in method.

We build a two-block Bayesian structural vector autoregression for each country. The first block is a Kilian and Murphy model of the global oil market. The second block contains domestic output, consumer prices, the bilateral exchange rate, and the policy rate. We identify the oil shocks with sign restrictions and with a bound on the short-run price elasticity of oil supply, in the spirit of Kilian and Murphy (2012, 2014). The elasticity bound disciplines the oil block. We also impose block exogeneity, so that domestic shocks do not move the global oil market, in the small-open-economy tradition of Cushman and Zha (1997). This is the natural assumption for small economies that take the world price of oil as given. We implement it in the near-vector-autoregression form, in which the oil block is identified on its own and then enters the domestic block as an exogenous regressor at the current month and at its lags. Doing so point-identifies the domestic transmission and sharpens the responses considerably relative to a joint model. A hard block-exogeneity restriction imposed instead as contemporaneous zeros interacts poorly with the importance-sampling identification that we use, which is why we adopt the near-vector-autoregression form, a point we return to in Section 3.6. With the transmission point-identified in this way, we focus on the differential responses across economies, and in particular on how a given kind of oil shock moves output differently in oil importers and oil exporters. Building on the unified treatment of conditional forecasting and structural scenarios in Antolín-Díaz, Petrella, and Rubio-Ramírez (2021), we then ask a sequence of scenario questions of the form, what happens to domestic output and inflation if the real price of oil rises by a given amount because of a particular shock. We grade the realism of every scenario with their Kullback and Leibler plausibility metric. The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 sets out the econometric framework, the identification, and the estimation. Section 4 reports results for eight economies across Asia, namely the five Southeast Asian economies of the Philippines, Malaysia, Thailand, Indonesia, and Vietnam, together with India, China, and South Korea. Section 5 concludes.

2 Related literature

Our paper draws on three strands of research, namely the structural analysis of the global oil market, the econometrics of conditional forecasts and scenarios in vector autoregressions, and the empirical work on how oil shocks reach individual economies.

The first strand begins with Hamilton (1983), who documented that almost every United States recession since the Second World War had been preceded by a sharp rise in the price of crude oil, and whose reading of the oil price as an exogenous driver shaped two

decades of empirical work surveyed in Kilian (2008). The modern treatment starts from Kilian (2009), who argued that the oil price is determined jointly with global quantities and that the macroeconomic consequences of an oil price change depend on the underlying shock, distinguishing supply shocks from aggregate demand and oil-specific demand shocks. Kilian and Murphy (2012) showed that sign restrictions alone are not enough to discipline this decomposition and that bounds on the short-run price elasticity of oil supply are needed, and Kilian and Murphy (2014) added above-ground inventories to separately identify speculative demand, giving the four-variable model that our oil block adopts. Baumeister and Hamilton (2019) reach related conclusions in a Bayesian framework that treats the identification as incomplete and makes the role of priors explicit, and Kilian (2019) defends the global activity index that these models use. Of particular relevance for Asia, Aastveit, Bjørnland, and Thorsrud (2015) estimate that demand from emerging economies, above all in Asia, accounts for more than twice as much of the variation in the real oil price as demand from developed economies, which motivates both our focus on the demand channel and our decision to let China feed back into the world oil market.

The second strand supplies our econometric machinery. The Bayesian treatment of vector autoregressions with informative priors descends from Sims and Zha (1998), the theory of identification with sign and zero restrictions is developed by Rubio-Ramírez, Waggoner, and Zha (2010) and Arias, Rubio-Ramírez, and Waggoner (2018), and narrative restrictions that tie identification to specific historical episodes are introduced by Antolín-Díaz and Rubio-Ramírez (2018). Conditional forecasting in identified models begins with Waggoner and Zha (1999), and Leeper and Zha (2003) introduced the idea that a policy intervention should be judged by whether the shocks it requires are modest. Building on both, Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) develop the structural scenario analysis that we apply, in which a future path for one variable is attributed to a chosen structural shock and the plausibility of the whole exercise is measured with a Kullback and Leibler statistic. Baumeister and Kilian (2014) pursue the closely related idea of grading oil price risk scenarios in real time. Two recent contributions bear directly on our scenario design. Kilian, Plante, Richter, and Zhou (2026) assess the inflationary impact of the 2026 Iran War by feeding oil price paths from a calibrated model of the global economy, one path for each assumed duration of the closure of the Strait of Hormuz, into a monthly VAR, and their tiered price paths anchor the Hormuz scenario in this paper. Hauzenberger and Pfarrhofer (2026) extend structural scenario analysis to high-dimensional Bayesian VARs and show, using the same tiered paths, that a given oil price trajectory is consistent with outcomes ranging from benign to pronounced stagflation depending on which structural shocks are permitted to deliver it, which is the within-country counterpart of the cross-country attribution result we

document. The block-exogeneity treatment of a small open economy that we use descends from Cushman and Zha (1997).

The third strand studies the transmission of oil shocks across countries. Peersman and Van Robays (2012) show for a set of industrialized economies that the output consequences of an oil price increase depend on the underlying shock and on each country's position in the energy trade, with net energy importers contracting after supply shocks while exporters are insulated or gain. For Asia specifically, Cuñado and Pérez de Gracia (2005) document significant effects of oil prices on activity and inflation in six Asian economies, Rafiq, Salim, and Bloch (2009) find a substantial role for oil price volatility in Thailand, and Basnet and Upadhyaya (2015) conclude that oil price shocks have mainly transitory effects in the founding members of the Association of Southeast Asian Nations. The study closest to ours is Cuñado, Jo, and Pérez de Gracia (2015), who identify the three Kilian and Murphy shocks for four large Asian oil consumers and find that responses depend strongly on the type of shock and that administered fuel prices blunt the pass-through in India and Indonesia. Relative to this work, we contribute a common framework applied uniformly to eight economies, with the same oil block, the same identification, and the same definition of every domestic variable, together with a scenario apparatus whose realism is graded rather than assumed. That uniformity is what allows the cross-country differences we report to be interpreted as economic rather than methodological.

3 Methodology

This section states the model and then reproduces the structural scenario framework of Antolín-Díaz, Petrella, and Rubio-Ramírez (2021). We keep their notation so that the reader can move between this paper and the original. After each block of equations we add a plain statement of what the equations mean.

3.1 The structural vector autoregression

Let $\mathbf{y}_t$ be the $n \times 1$ vector of observable variables. The structural vector autoregression with p lags is written in equation (1),

$$\mathbf{y}_t' \mathbf{A}_0 = \mathbf{x}_t' \mathbf{A}_+ + \boldsymbol{\varepsilon}_t', \qquad t = 1, \ldots, T, \tag{1}$$

where $\mathbf{A}_+' = [\, \mathbf{A}_1' \;\; \cdots \;\; \mathbf{A}_p' \;\; \mathbf{d}' \,]$ and $\mathbf{x}_t' = [\, \mathbf{y}_{t-1}' \;\; \cdots \;\; \mathbf{y}_{t-p}' \;\; 1 \,]$. The matrix $\mathbf{A}_0$ collects the contemporaneous structural coefficients and is assumed invertible. The vector $\boldsymbol{\varepsilon}_t$ collects the structural shocks and, conditional on the past, is distributed as $N(\mathbf{0}_{n\times 1}, \mathbf{I}_n)$. The structural

shocks are mutually orthogonal and carry an economic interpretation. The reduced form that corresponds to equation (1) is given in equation (2),

$$\mathbf{y}_t' = \mathbf{x}_t' \mathbf{B} + \mathbf{u}_t', \qquad \mathbf{B} = \mathbf{A}_+ \mathbf{A}_0^{-1}, \quad \mathbf{u}_t' = \boldsymbol{\varepsilon}_t' \mathbf{A}_0^{-1}, \quad \boldsymbol{\Sigma} = \mathbb{E}[\mathbf{u}_t \mathbf{u}_t'] = (\mathbf{A}_0 \mathbf{A}_0')^{-1}. \tag{2}$$

The reduced-form innovations $\mathbf{u}_t$ are correlated and have no direct interpretation, whereas the structural shocks $\boldsymbol{\varepsilon}_t$ are orthogonal and interpretable. As is well known, equation (1) is not identified from the data alone. Following Rubio-Ramírez, Waggoner, and Zha (2010), one reparameterizes the model in terms of the reduced-form objects $\mathbf{B}$ and $\boldsymbol{\Sigma}$ together with an $n \times n$ orthogonal rotation matrix $\mathbf{Q}$. A given choice of $\mathbf{Q}$ selects one observationally equivalent set of structural parameters. Identification therefore amounts to restricting the set of admissible rotations $\mathbf{Q}$.

3.2 The unconditional forecast

Consider forecasting the observables h periods ahead. Stack the forecasts as $\mathbf{y}_{T+1,T+h} = [\mathbf{y}_{T+1}' \ \cdots \ \mathbf{y}_{T+h}']'$ and the future shocks as $\boldsymbol{\varepsilon}_{T+1,T+h}$. The forecast can be written as in equation (3),

$$\mathbf{y}_{T+1,T+h} = \mathbf{b}_{T+1,T+h} + \mathbf{M}' \boldsymbol{\varepsilon}_{T+1,T+h}, \tag{3}$$

where the vector $\mathbf{b}_{T+1,T+h}$ is predetermined and depends only on the history of the observables and on the reduced-form parameters, and where the matrix $\mathbf{M}$ collects the structural moving-average coefficients that map the future shocks into the future observables. It follows that the unconditional forecast is distributed as in equation (4),

$$\mathbf{y}_{T+1,T+h} \sim N(\mathbf{b}_{T+1,T+h}, \ \mathbf{M}'\mathbf{M}) \,. \tag{4}$$

While $\mathbf{M}$ depends on the structural parameters through the choice of $\mathbf{Q}$, the product $\mathbf{M}'\mathbf{M}$ depends only on the reduced-form parameters. In the absence of any further information the future shocks keep their unconditional distribution, which is $\boldsymbol{\varepsilon}_{T+1,T+h} \sim N(\mathbf{0}_{nh\times 1}, \mathbf{I}_{nh})$. In words, the unconditional forecast is the model's best guess for the future when no scenario is imposed, and its central path $\mathbf{b}_{T+1,T+h}$ can be computed without taking a stand on identification.

3.3 The general framework for conditional forecasts and structural scenarios

A scenario is a set of linear restrictions on the future observables. In its most general form the restriction is written as in equation (5),

$$\mathbf{C}\,\mathbf{y}_{T+1,T+h} \sim N(\mathbf{f}_{T+1,T+h},\ \boldsymbol{\Omega}_f)\,, \tag{5}$$

where $\mathbf{C}$ is a $k \times nh$ matrix that selects and combines the future observables, $\mathbf{f}_{T+1,T+h}$ is the target for those combinations, and $\boldsymbol{\Omega}_f$ is the covariance matrix of the target. Substituting equation (3) into equation (5) turns a restriction on the observables into a restriction on the future shocks. Writing the mean and the variance of the shocks under the scenario as $\boldsymbol{\mu}_\varepsilon$ and $\boldsymbol{\Sigma}_\varepsilon$, and defining $\mathbf{D} = \mathbf{C}\mathbf{M}'$, the restriction requires that $\mathbf{D}\boldsymbol{\mu}_\varepsilon = \mathbf{f}_{T+1,T+h} - \mathbf{C}\mathbf{b}_{T+1,T+h}$ and that $\mathbf{D}\boldsymbol{\Sigma}_\varepsilon\mathbf{D}' = \boldsymbol{\Omega}_f$.

When the number of restrictions k is smaller than nh, the system has many solutions. Following Penrose, Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) select the solution that stays as close as possible to the unconditional distribution of the shocks. That solution is given in equations (6) and (7),

$$\boldsymbol{\mu}_\varepsilon = \mathbf{D}^\star\left(\mathbf{f}_{T+1,T+h} - \mathbf{C}\mathbf{b}_{T+1,T+h}\right), \tag{6}$$

$$\boldsymbol{\Sigma}_\varepsilon = \mathbf{I}_{nh} - \mathbf{D}^\star\mathbf{D} + \mathbf{D}^\star\boldsymbol{\Omega}_f\,\mathbf{D}^{\star\prime}, \tag{7}$$

where $\mathbf{D}^\star = \mathbf{D}'(\mathbf{D}\mathbf{D}')^{-1}$ is the Moore and Penrose generalized inverse of $\mathbf{D}$. This choice minimizes the deviation of the mean and of the covariance of the future shocks from their unconditional values, measured by the Frobenius norm. The plain reading is that the model bends the shocks by the least amount that is needed to satisfy the scenario, and it leaves everything that the scenario does not pin down at its unconditional value.

Mapping the shock distribution back into the observables through equation (3) gives the restricted forecast. Its mean is given in equation (8) and its covariance in equation (9),

$$\mathbf{y}^{\text{scenario}}_{T+1,T+h} = \mathbf{b}_{T+1,T+h} + \mathbf{M}'\mathbf{D}^\star\left(\mathbf{f}_{T+1,T+h} - \mathbf{C}\mathbf{b}_{T+1,T+h}\right), \tag{8}$$

$$\boldsymbol{\Sigma}_y = \mathbf{M}'\mathbf{M} + \mathbf{M}'\mathbf{D}^\star\left(\boldsymbol{\Omega}_f - \mathbf{D}\mathbf{D}'\right)\mathbf{D}^{\star\prime}\mathbf{M}. \tag{9}$$

Equation (8) says that the scenario updates the unconditional central path by an amount that depends on how far the target $\mathbf{f}_{T+1,T+h}$ lies from what the model expected, which is $\mathbf{C}\mathbf{b}_{T+1,T+h}$. Equation (9) shows that the choice $\boldsymbol{\Omega}_f = \mathbf{D}\mathbf{D}'$ makes the second term vanish and leaves $\boldsymbol{\Sigma}_y = \mathbf{M}'\mathbf{M}$. In that case the scenario shifts the central path without shrinking

the forecast bands. Preserving the bands in this way is what makes the method suitable for density forecasting and risk assessment, rather than only for point projection. Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) further prove that this solution is the one that minimizes the Kullback and Leibler divergence from the unconditional forecast, which connects the method to entropic tilting in the tradition of the conditional forecasting literature of Waggoner and Zha (1999).

3.4 Three special cases and the structural scenario

Three special cases sit inside this framework. In the first case, which is the classic conditional forecast of Waggoner and Zha (1999), the matrix $\mathbf{C}$ simply selects a subset of the future observables and holds them on a path. Here $\mathbf{C}$ does not depend on the identification and the exercise is silent about which shocks deliver the path. In the second case one restricts the path of a subset of the structural shocks. Because the shocks can be recovered from the observables through $\boldsymbol{\varepsilon}_{T+1,T+h} = (\mathbf{M}')^{-1}(\mathbf{y}_{T+1,T+h} - \mathbf{b}_{T+1,T+h})$, a restriction on the shocks is again a linear restriction on the observables, now with a selection matrix that depends on identification.

The third case is the structural scenario, which is the tool we use. It combines a path for one or more observables with the requirement that only a chosen subset of shocks, called the driving shocks, may depart from the unconditional distribution. The remaining shocks, called the non-driving shocks, are held at their unconditional distribution. Let $\boldsymbol{\Xi}$ be the $k_s \times nh$ selection matrix that picks out the non-driving shocks. The structural scenario stacks the two requirements as in equation (10),

$$\begin{bmatrix} \mathbf{C} \\ \boldsymbol{\Xi}(\mathbf{M}')^{-1} \end{bmatrix} \mathbf{y}_{T+1,T+h} \sim N\left(\begin{bmatrix} \mathbf{f}_{T+1,T+h} \\ \boldsymbol{\Xi}(\mathbf{M}')^{-1}\mathbf{b}_{T+1,T+h} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Omega}_f & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{k_s} \end{bmatrix} \right). \tag{10}$$

The upper block states that a selection of variables must follow the path $\mathbf{f}_{T+1,T+h}$ in expectation. The lower block states that the non-driving shocks must keep the unconditional standard normal distribution. Equation (10) is again a special case of equation (5), so the closed-form solutions in equations (6) through (9) apply directly. A key point is that the stacked selection matrix now depends on $\mathbf{M}$, and therefore on the choice of $\mathbf{Q}$, so a structural scenario requires the driving shocks to be identified. When the restricted variable is a policy instrument and the single driving shock is the associated policy shock, the exercise reduces to the modest policy intervention of Leeper and Zha (2003).

3.5 How plausible is a scenario

Any structural scenario forces the driving shocks away from their unconditional standard normal distribution. When that force is large, the scenario asks the model to do something it rarely does in the data, and the results become fragile in the sense of the Lucas critique. To measure this, Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) use the Kullback and Leibler divergence between the distribution of the shocks under the scenario, which we call N_{SS}, and their unconditional distribution, which we call N_{UF} and which is standard normal. That divergence is given in equation (11),

$$D_{KL}(N_{SS} \,\|\, N_{UF}) = \tfrac{1}{2}\left[\, \boldsymbol{\mu}_{\varepsilon}'\boldsymbol{\mu}_{\varepsilon} + \operatorname{tr}(\boldsymbol{\Sigma}_{\varepsilon}) - nh - \ln|\boldsymbol{\Sigma}_{\varepsilon}| \,\right], \tag{11}$$

where $\boldsymbol{\mu}_{\varepsilon}$ and $\boldsymbol{\Sigma}_{\varepsilon}$ come from equations (6) and (7). The divergence is easy to compute but hard to read on its own. To make it interpretable, Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) calibrate it to a number q that compares a fair coin with a biased coin, as in equation (12),

$$q = \tfrac{1}{2}\left(1 + \sqrt{1 - e^{-2z/(nh)}}\right), \qquad z = D_{KL}(N_{SS} \,\|\, N_{UF})\,. \tag{12}$$

A value of q equal to one half describes a scenario that is indistinguishable from the unconditional forecast. A value close to one describes an intervention that is very unusual by historical standards. We report q for every scenario, and we treat scenarios with a high value of q with caution.

Three properties of q matter for reading our results. First, its scale runs from one half to one, not from zero to one. The floor of one half is reached exactly when the scenario requires no unusual shocks at all, so a value such as 0.8 sits well above the midpoint of the possible range and describes a strong intervention. Second, there is no formal threshold that separates a plausible scenario from an implausible one. The statistic is a continuous, comparative measure, and the closest thing to a rule of thumb is the modesty criterion of Leeper and Zha (2003), under which an intervention is modest when the shocks it requires stay within the range the model has seen historically. We therefore use q to rank scenarios against one another and against the floor of one half, not to accept or reject them. Third, q depends on the whole path that a scenario imposes, not only on its peak. Holding the oil price at an elevated level for many months requires a sustained sequence of same-signed shocks and yields a much higher q than a single shock whose price effect decays at the model's own speed. For every scenario in this paper we compute q for the exact price path that the reported responses embody, namely the path implied by the single scaled shock, and where instructive we also report the q of the corresponding sustained peg to show how

much plausibility the persistence itself consumes.

3.6 Empirical specification and identification

We now specialize the framework to our application. The vector $\mathbf{y}_t$ has eight elements and is partitioned into an oil block and a domestic block. The oil block follows Kilian and Murphy (2014) and contains the percentage change in global crude-oil production, a measure of global real economic activity, the real price of oil, and the change in global crude-oil inventories. The domestic block contains the log of manufacturing output, the log of the consumer price index, the log of the bilateral exchange rate against the United States dollar, and the short-term policy interest rate.

Identification rests on three elements. First, we identify three oil shocks with the Kilian and Murphy sign pattern on impact. A flow-supply shock lowers production, lowers global activity, and raises the real price. A flow-demand shock raises production, activity, and the real price. A speculative demand shock raises the real price and inventories while activity is weak. The sign restrictions are implemented with the algorithm of Arias, Rubio-Ramírez, and Waggoner (2018). We impose the sign pattern on impact only. Kilian and Murphy (2014) additionally require that the production decline and the price increase that follow a flow-supply disruption persist for a full year. We do not adopt these dynamic restrictions, because within the importance sampler the additional inequality constraints, two variables at each of twelve further horizons, reduce the acceptance rate so severely that the sampler fails to deliver a usable posterior in reasonable time. The discipline that the dynamic restrictions are designed to provide is instead supplied by the elasticity bound that follows, which Kilian and Murphy (2012) identify as the essential ingredient. Second, we bound the impact price elasticity of oil supply to a small value, because Kilian and Murphy (2012) show that sign restrictions on their own admit implausibly large elasticities and therefore explosive dynamics. We implement this bound as a rejection filter on the posterior draws, keeping only draws whose implied short-run supply elasticity is small. This is the step that gives the oil shocks an economically sensible role, and it does so without any zero restriction. Third, for the policy scenarios we also sign-identify a domestic monetary-policy shock, under which a contractionary shock raises the policy rate and lowers output, prices, and the value of the currency on impact. Only the shocks that drive a given scenario need to be identified, so the remaining domestic shocks are left set-identified. Where useful, narrative restrictions in the manner of Antolín-Díaz and Rubio-Ramírez (2018) can be added around salient episodes as a robustness check.

Fourth, we impose block exogeneity, so that domestic shocks have no effect on the global

oil block. A small open economy takes the world price of oil as given, so this is the economically natural restriction. We impose it in the near-vector-autoregression form of Cushman and Zha (1997). The oil block is estimated and identified on its own as described above. The domestic block is then estimated as a reduced-form vector autoregression in which the four oil variables enter as exogenous regressors, at the current month and at six lags, so that oil affects the domestic economy while the domestic economy does not feed back into the oil block. The domestic response to an identified oil shock is obtained by propagating the identified oil-block path through the domestic coefficients. This point-identifies the transmission and delivers much tighter credible bands than the joint model. We do not instead impose block exogeneity as contemporaneous zero restrictions on the impact matrix, because within the importance sampling that identifies the sign restricted model those zeros make the effective sample size collapse toward one, so that a single draw dominates the posterior and the reported bands would be spurious. The near-vector-autoregression form avoids this problem and encodes the same economic assumption. Table 1 summarizes the full identification scheme.

3.7 Estimation and data

We estimate the model by Bayesian methods with a Minnesota-style prior on the reduced-form parameters, in the tradition of Sims and Zha (1998), at monthly frequency with six lags. Posterior draws of the structural parameters that satisfy the sign restrictions are obtained with the algorithm of Arias, Rubio-Ramírez, and Waggoner (2018), and we then apply the elasticity filter to those draws.

The global oil block is common to all countries. Global crude-oil production and the change in inventories come from the United States Energy Information Administration. The real price of oil is the Brent spot price deflated by the United States consumer price index, both taken from the Federal Reserve Economic Data service. Global real activity is the index of Kilian (2009), corrected as in Kilian (2019) and published by the Federal Reserve Bank of Dallas. The domestic series come from the International Financial Statistics of the International Monetary Fund and from the World Bank. Consumer prices, the bilateral exchange rate, and the policy rate are taken from the International Financial Statistics for every economy. Output is the monthly index of industrial production from the World Bank Global Economic Monitor, again for every economy, so the cross-country comparison rests on a single definition of activity. Table 2 collects the variables, their sources, the series codes, and the coverage, and descriptive statistics by economy appear in Appendix A.

Table 1: Summary of the identification scheme

Panel A. Sign restrictions on the impulse responses of the global oil block

	Structural shock		
Response of	Flow supply, ε_t^s	Flow demand, ε_t^d	Speculative demand, ε_t^{sd}
Global crude-oil production	$-$	$+$	$+$
Global real activity	$-$	$+$	$-$
Real price of oil	$+$	$+$	$+$
Change in oil inventories	unrestricted	unrestricted	$+$
Horizons restricted	impact	impact	impact

Panel B. Restrictions beyond the sign pattern

Restriction	Statement
Supply elasticity bound	The impact price elasticity of oil supply implied by each posterior draw must lie below 0.05. The bound is relaxed stepwise to at most 0.15 when fewer than sixty draws satisfy it. The bound follows Kilian and Murphy (2012).
Block exogeneity	The four oil variables enter each domestic block as exogenous regressors at the current month and at six lags, and no domestic variable enters the oil block. The restriction follows Cushman and Zha (1997) and point-identifies the domestic transmission.
Treatment of China	China alone is estimated in the joint eight-variable system, so that feedback from the Chinese economy to the world oil market is permitted at all horizons.

Notes. In Panel A, an entry of $+$ restricts the response of the row variable to the column shock to be nonnegative over the restricted horizons, an entry of $-$ restricts it to be nonpositive, and an unrestricted entry leaves the response free. Each shock is normalized so that it raises the real price of oil. The sign pattern is that of Kilian and Murphy (2014) and is imposed with the algorithm of Arias, Rubio-Ramírez, and Waggoner (2018). *Source.* Authors' summary.

Table 2: Variables, sources, and coverage

Variable	Description	Source and code	Coverage
Global oil block (common to all economies)			
Oil production	World crude oil production, log change	EIA STEO, COPR_WORLD.M	1993 onward
Global activity	Index of global real economic activity	FRED, IGREA	1968 onward
Real oil price	Brent spot deflated by US CPI	FRED, POILBREUSDM, CPIAUCSL	1992 onward
Oil inventories	OECD petroleum stocks, change	EIA, INTL.5-5-OECD-MBBL.M	1980 onward
Domestic block (each economy, ISO code in the series)			
Industrial production	Output index, seasonally adjusted	World Bank GEM, M-IPTOTSAKD-ISO	1991 onward
Consumer prices	Consumer price index	IMF IFS, M.ISO.PCPI_IX	varies
Exchange rate	Local currency per US dollar, average	IMF IFS, M.ISO.ENDA_XDC_USD_RATE	varies
Interest rate	Policy or money-market rate	IMF IFS, see note	varies

Notes. The interest rate is the policy rate (IFS FPOLM_PA) for South Korea and the Southeast Asian economies, the one-year benchmark lending rate (FILR_PA) for China, and the call money rate (FIMM_PA) for India, chosen in each case for coverage. Estimation samples begin in 2001 where the data allow. *Source.* Authors' compilation.

4 Results

We report results for eight economies across Asia, namely the five Southeast Asian economies of the Philippines, Malaysia, Thailand, Indonesia, and Vietnam, together with the three large Asian economies of India, China, and South Korea. All eight models use the same oil block, the same identification, and the same industrial production series from the World Bank Global Economic Monitor. We estimate each over a common window that begins in 2001, which keeps the 1997 and 1998 Asian financial crisis out of the sample. Two economies have shorter windows because of data availability, namely Vietnam, for which the output index begins in 2008, and India, for which the money-market rate that we use as the policy instrument is continuous only from 2006 and the domestic series end in 2022.

We identify three oil shocks in the manner of Kilian and Murphy, namely a flow-supply shock, a flow-demand shock, and a speculative demand shock, and we trace each of them through to domestic output and prices with the identification of Section 3. For China, which is large enough to influence the world oil market, we allow the domestic economy to feed back into the oil block, as discussed below. The results that follow are read shock by shock and country by country.

4.1 Oil shocks are a material driver of the regional business cycle

That oil shocks matter for Asian economies is well established. Cuñado, Jo, and Pérez de Gracia (2015) and Cuñado and Pérez de Gracia (2005) document sizable effects of oil price shocks on activity and inflation across a range of Asian economies, Basnet and Upadhyaya (2015) find similar effects for the larger members of the Association of Southeast Asian Nations, and Rafiq, Salim, and Bloch (2009) report a substantial role for oil price volatility in Thailand. Our decomposition is consistent with this literature. Figure 4 reports, for each economy, the share of the two-year forecast-error variance of output that the global oil market accounts for. The share is large, running from about 27 percent for the least oil-sensitive economy to about 63 percent for the most, and the variance of consumer prices behaves similarly. Because we treat the oil market as exogenous, this share carries the full comovement between oil and the domestic economy and is best read as an upper bound. Even read that way the message is clear, since in every economy oil is a first-order driver of the domestic business cycle. The more interesting question, which the rest of this section takes up, is not how much oil matters but how it matters, since a given movement in the oil price can come from very different shocks.

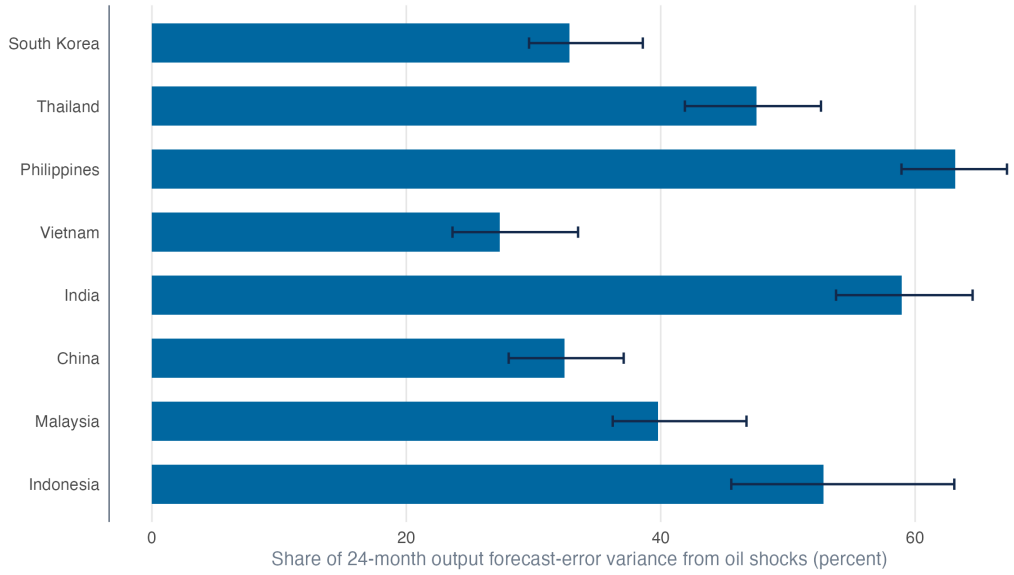


Figure 4: The global oil block accounts for a material share of output variance in every economy

Notes. Posterior median share of the 24-month forecast-error variance of output attributable to the global oil block, by economy, with the 68 percent credible band. *Source.* Authors' structural VAR.

4.2 The source of the oil price change is what matters

Consider the two shocks that raise the price of oil. Both push the price up, but they are otherwise opposite events. A flow-supply shock is a disruption to the physical availability of oil, such as an outage at a large producer or a geopolitical threat to shipping. It raises the price precisely because it makes oil scarcer, and it does so while global production and global economic activity are falling. For an economy that buys oil abroad, it is a pure increase in the cost of an essential import that arrives with no offsetting benefit, so it acts like an adverse terms-of-trade shock. A flow-demand shock is the reverse. It is a broad strengthening of global demand for industrial commodities, the kind that accompanies a synchronized world expansion, and it raises the oil price as a by-product of a booming world economy. The higher price now comes bundled with stronger external demand, firmer commodity-linked incomes, and more buoyant trade. Whether a given oil price increase is contractionary or expansionary for a domestic economy therefore depends entirely on which of these two shocks is behind it.

Figure 5 confirms that the distinction is what the data see. It reports, for each economy, the average output response over two years to a one-standard-deviation oil-supply shock and to a global-demand shock, each with a 68 percent credible band. The clearest message is about the demand shock. A global-demand-driven increase in the oil price raises domestic output, and the effect is positive and statistically distinguishable from zero in six of the eight

economies, at +2.14 percent for the Philippines, +0.50 for Malaysia, +0.71 for Thailand, +0.78 for Indonesia, +0.66 for India, and +0.44 for South Korea, with only Vietnam and China close to zero. The demand tailwind dominates the energy-cost drag for these open, trade-intensive economies. The supply shock, examined next, is a very different matter. It is not the oil price change on its own that matters for the domestic economy but the shock behind it.

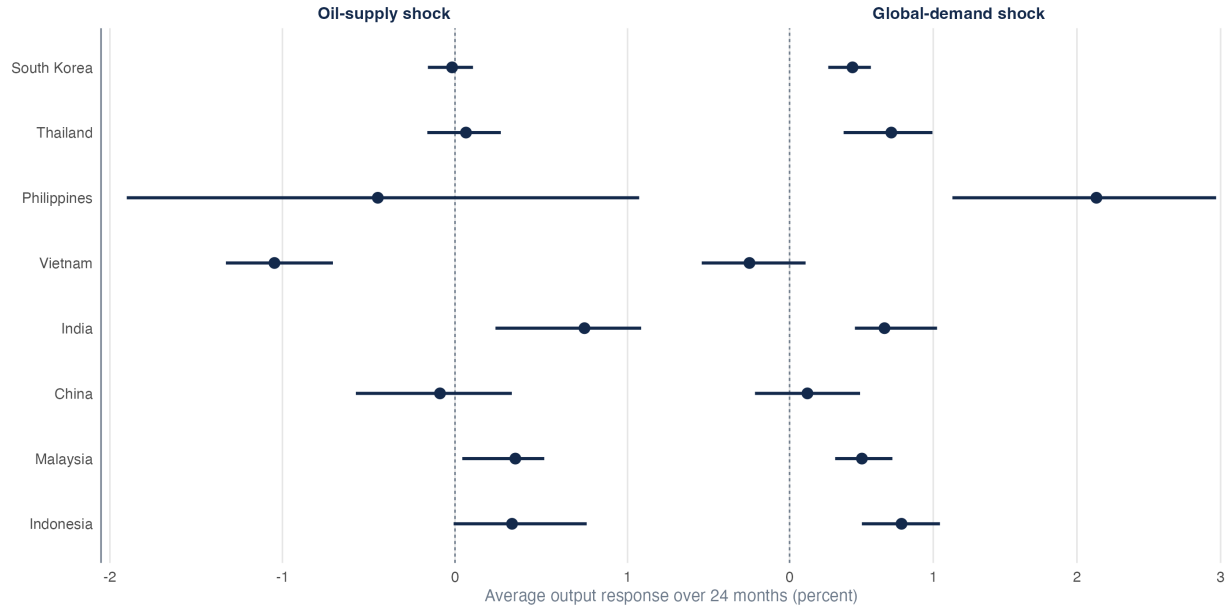


Figure 5: A demand-driven oil increase raises output, while the supply response is heterogeneous

Notes. Posterior median and 68 percent credible band of the average output response over the 24 months following a one-standard-deviation price-increasing oil shock, by economy. Economies are ordered from most energy-import-intensive at the top to most export-intensive at the bottom. *Source.* Authors' structural VAR.

4.3 The supply channel is heterogeneous across economies

The supply shock exhibits greater heterogeneity across economies, and it is here that the country dimension is most informative. A price-raising oil-supply shock lowers output most clearly in Vietnam, at -1.05 percent with a band that excludes zero, and points negative in the Philippines at -0.45 and in China at -0.09. It is small and close to zero in South Korea at -0.02 and in Thailand at +0.06, and positive in Malaysia at +0.35, in Indonesia at +0.33, and in India at +0.75. The responses do not line up cleanly with each economy's position in the energy trade, since India in particular is an oil importer yet shows a positive response. What the eight economies share is that the supply response is small in absolute size, and much smaller than the demand response, which is consistent with the view of Kilian and

Murphy (2012) that flow-supply shocks are a modest part of what moves the oil price. The differential across economies, rather than any single number, is the substantive result, and it cautions against reading a common oil-supply story onto a diverse region.

China. China is a special case, because it is large enough to move the world oil market rather than simply take the oil price as given. We therefore estimate China with the feedback from its own economy to the oil market allowed, and it is these estimates that appear for China in Figure 5 and throughout. Once the feedback is permitted, China behaves like a conventional oil importer, with an output response of -0.09 percent to a supply shock and +0.12 percent to a demand shock, both small and imprecisely estimated. The point is that China is too large for the small-open-economy reading that suits the other seven economies, and once that is recognized its results are unremarkable.

4.4 What the impulse responses do and do not show

Figures 6 and 7 show the full set of impulse responses for the Philippines, an oil importer, and for Malaysia, an economy that is close to energy balance. They illustrate the lessons above. The response that is visible and consistent is the response to the global-demand shock, which raises consumer prices in both economies and tends to raise output as well, the positive comovement of a shock that reflects a stronger world economy. The responses to the pure oil-supply and speculative demand shocks are muted and their bands are wide. The full set of impulse responses for all eight economies is in Appendix C.

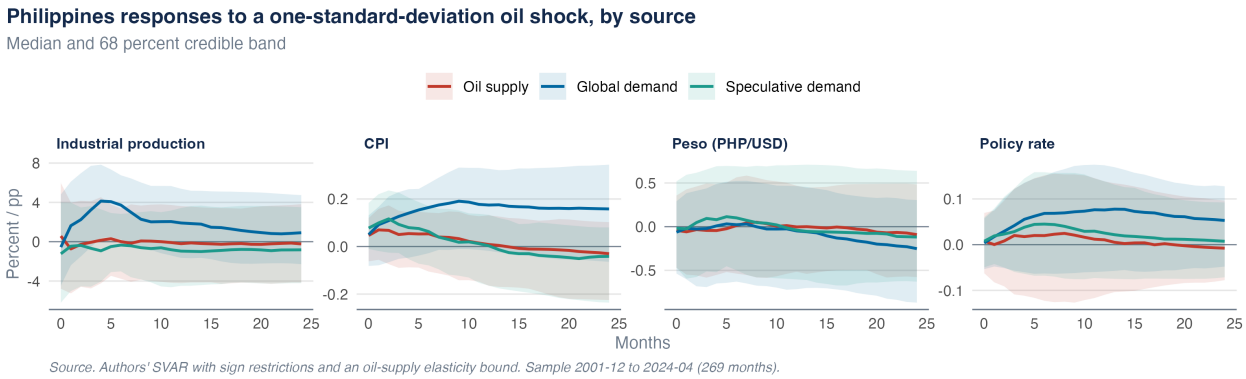


Figure 6: Philippine responses to a one-standard-deviation oil shock, sign-restricted identification

4.5 The same oil price, two different sources

The structural scenario framework of Section 3 lets us state the central lesson of the paper as a single controlled experiment. We ask what happens to an economy when the real price

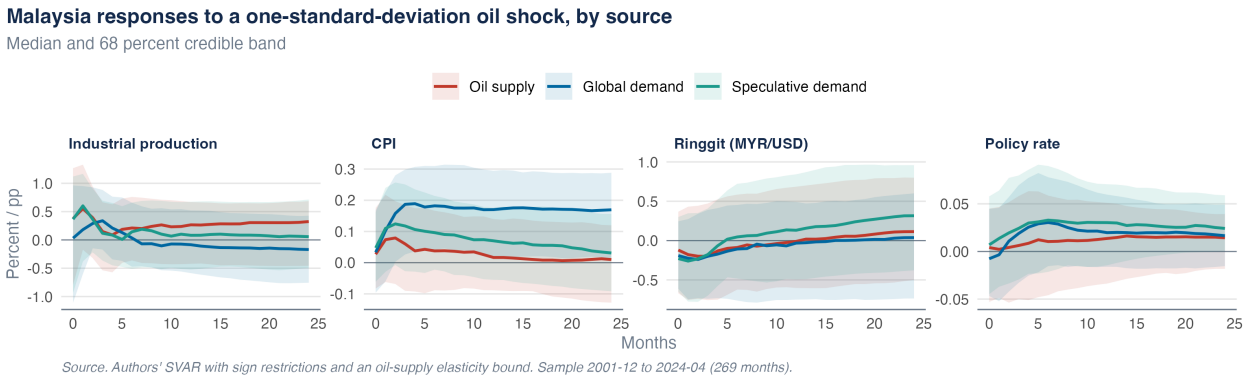


Figure 7: Malaysian responses to a one-standard-deviation oil shock, sign-restricted identification

of oil rises by twenty percent, and we ask the question twice, once when the increase is generated by an oil-supply shock and once when it is generated by a global-demand shock. Because the two experiments condition on the same movement in the oil price and differ only in the shock that produces it, any difference in the domestic outcome is due to the source of the shock and not to its size. We illustrate the experiment with the Philippines in Figure 8 and Table 3, and then show that the same pattern holds across all eight economies in Figure 9.

The contrast is instructive. A demand-driven increase leaves Philippine output higher after two years, by +0.50 percent, because the stronger world economy that produces the higher price also lifts external demand. A supply-driven increase of the same size leaves output lower, by -2.25 percent, the classic adverse terms-of-trade effect for an oil importer. Consumer prices rise by +0.72 percent in the demand case and are little changed in the supply case, at -0.34 percent. The same twenty percent rise in the oil price therefore carries opposite implications for output depending on its cause.

The plausibility statistic q is computed for the exact experiment that the figure shows, namely the oil price path implied by the single shock, which rises to its peak and then decays at the model's own speed. Graded this way, q is 0.76 for the demand case and 0.87 for the supply case. Recall from Section 3 that the scale of q runs from one half, a scenario requiring no unusual shocks, to one, and that no formal threshold separates plausible from implausible. Read against that scale, the demand experiment is the more ordinary of the two, which accords with the historical decomposition in Figure 2, where demand shocks account for most large oil price movements. The persistence of a scenario is itself expensive in plausibility. Had we instead held the oil price at its peak for a full year, the same twenty percent increase would carry a q of 0.81 in the demand case and 0.93 in the supply case, because a sustained peg requires a long sequence of same-signed shocks that the model rarely produces. This distinction matters when comparing our scenarios with each other and with

the larger episodes considered below.

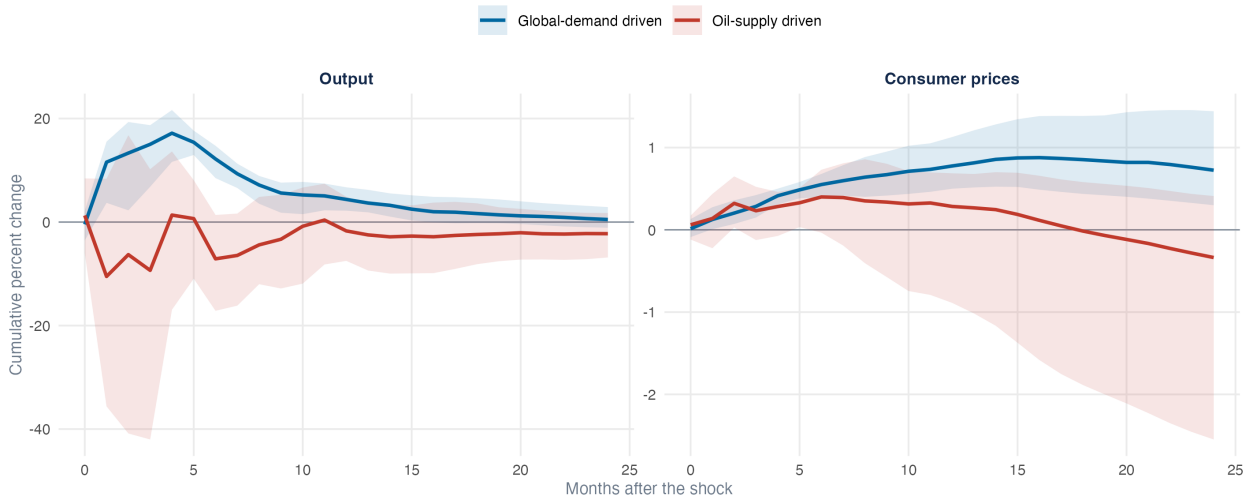


Figure 8: The same twenty percent oil price increase, from two different sources, for the Philippines

Notes. Cumulative response of Philippine output and consumer prices to a single oil shock that raises the real oil price twenty percent at its peak, median and 68 percent band. Blue conditions the increase on a global-demand shock, red on an oil-supply shock. *Source.* Authors' structural VAR.

Table 3: Philippine outcomes after two years, same twenty percent oil price increase

	Demand-driven	Supply-driven
Output (cumulative percent)	+0.50	-2.25
Consumer prices (cumulative percent)	+0.72	-0.34
Plausibility statistic q	0.76	0.87

The Philippine experiment is not special. Figure 9 repeats it for all eight economies, plotting the output response to the same twenty percent increase in the real oil price under the two drivers. The demand-driven path, in blue, lies above the supply-driven path, in red, in almost every economy, which is the visual signature of the central result. The gap between the two paths is the part of the output response that is due to the source of the oil price change rather than to its size, and it is present across the region even though its width varies from one economy to the next. In the oil importers the supply-driven path is generally the more damaging of the two, while in the net exporters the two paths are closer together and both tend to be positive.

4.6 A regional scenario, the global demand surge of 2007 and 2008

The Philippine experiment isolates the driver for a single economy. A complementary experiment holds the driver fixed and varies the economy, which is the natural way to gauge

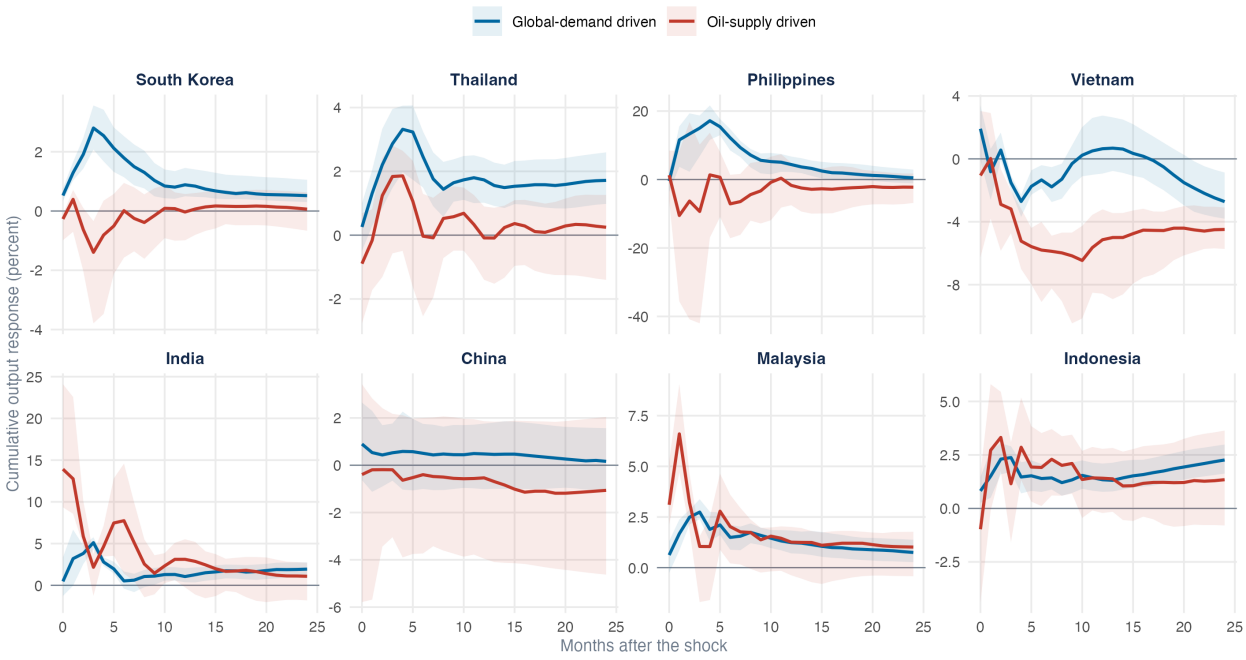


Figure 9: The same oil price increase from two sources, output responses across the region

Notes. Cumulative output response to a single oil shock that raises the real oil price twenty percent at its peak, for each economy, median and 68 percent band. Blue conditions the increase on a global-demand shock and red on an oil-supply shock. *Source.* Authors' structural VAR.

the regional consequences of a common shock. For that shock we use a concrete historical episode. During 2007 and the first half of 2008 the real price of oil roughly doubled, and the decomposition in Figure 2 attributes that run-up overwhelmingly to global demand, as the world economy, and emerging Asia in particular, expanded rapidly. We represent the episode as a global-demand shock that raises the real oil price by forty percent at its peak, and we apply it to all eight economies. Figure 10 reports the cumulative response of output and consumer prices after two years.

Because the demand channel is the well-identified one, the regional responses are coherent and, for most economies, precisely estimated. Output rises significantly in five of the eight economies, by +4.5 percent in Indonesia, +3.9 in India, +3.4 in Thailand, +1.5 in Malaysia, and +1.0 in South Korea. It also rises, though less precisely, in the Philippines at +1.0 percent, while China is little changed at +0.3 percent. Vietnam is the exception, with output significantly lower at -5.4 percent, consistent with its atypical response to the demand shock noted above. Consumer prices rise throughout, since the higher oil price feeds into domestic costs while the world economy is strong. A demand-driven oil surge of the kind seen in 2008 is therefore, on balance, expansionary for these open Asian economies, even for the many that import most of their oil, because the global boom that drives the oil price also drives

their exports. The scenario is a large one and its plausibility statistic reflects that, at 0.91 for the Philippines and similar values elsewhere, yet it remains below the ceiling that the supply-driven counterpart below attains, which accords with the fact that the episode it mimics actually happened.

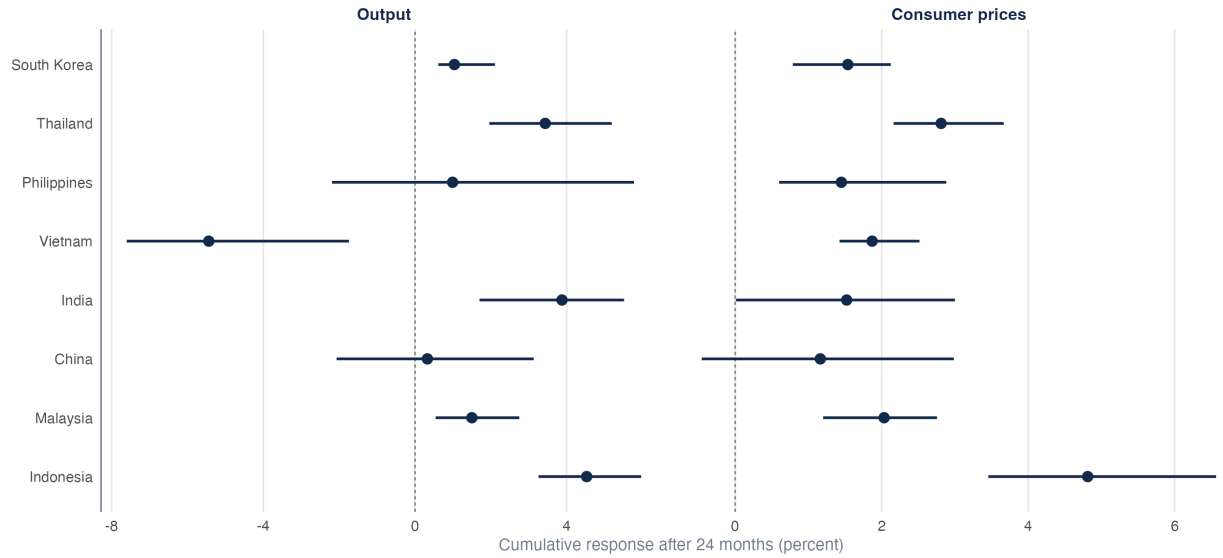


Figure 10: A global demand surge of the kind seen in 2007 and 2008, responses across the region

Notes. Cumulative response of output and consumer prices after 24 months to a global-demand shock that raises the real oil price forty percent at its peak, by economy, median and 68 percent band. Economies are ordered from most energy-import-intensive to most export-intensive. *Source.* Authors' structural VAR.

4.7 A supply-driven counterpart, a closure of the Strait of Hormuz

The natural supply-driven counterpart is a disruption in the Persian Gulf. About a fifth of the world's seaborne crude oil, together with a large share of liquefied natural gas, passes through the Strait of Hormuz, and the strait was in fact closed during the 2026 Iran War. Rather than invent a price path, we anchor the scenario in the analysis of that event by Kilian, Plante, Richter, and Zhou (2026), who calibrate the oil price consequences of closures of increasing duration in a model of the global economy. Their projections imply that a closure removing about twenty percent of world oil supplies raises the price of crude from a sixty dollar baseline to a peak of 103, 132, or 167 dollars per barrel as the closure extends from one to two to three quarters. We translate these peaks into three tiers for the real oil price, and we implement each tier as an oil-supply shock scaled to the corresponding peak, applied identically to all eight economies. Table 4 states the design and Figure 11 reports the imposed price paths and the output outcomes.

Table 4: Design of the Strait of Hormuz tiers

	One-quarter closure	Two-quarter closure	Three-quarter closure
Peak WTI price, dollars	103	132	167
Peak rise over baseline	+72 percent	+120 percent	+178 percent
Peak in log points	54	79	102
Plausibility q	1.00	1.00	1.00

Notes. Price peaks follow Kilian, Plante, Richter, and Zhou (2026), relative to their sixty dollar pre-war baseline. Each tier is a single flow-supply shock scaled to the stated peak, with the price then decaying at the model's own speed. The plausibility statistic is computed on the Philippine oil block. *Source.* Authors' summary.

The tiers turn a single stylized number into a dose-response reading, and the response is steepest exactly where the supply channel is sharpest. Vietnamese output after two years is lower by -12.2 percent under a one-quarter closure, -17.7 under two quarters, and -23.0 under three, the one gradient in the region that is estimated precisely throughout. The Philippine medians fall from -6.1 to -11.5 percent across the tiers and China's from -2.9 to -5.4, though with bands that include zero, while the net exporters Indonesia and Malaysia gain modestly at every tier and South Korea and Thailand barely move. The ordering of economies is unchanged across tiers, since the model is linear, so what the tiers add is the scale of the exposure and a bridge to the event-based projections of Kilian, Plante, Richter, and Zhou (2026).

Two cautions temper the reading. First, every tier sits essentially at the ceiling of the plausibility scale, as Table 4 records, because a price spike of this size attributed purely to the flow-supply shock is far outside the sample experience of that shock. Second, and reinforcing the first, Hauzenberger and Pfarrhofer (2026) show with the same tiered paths that the attribution itself is what carries the macroeconomic content. In their large United States model the same oil price trajectory produces a benign episode when its structural origins are left unrestricted and pronounced stagflation, with much wider predictive bands, when it is attributed to the oil supply shock alone. Our wide bands under the supply attribution are the cross-country counterpart of that finding. Certainty about the structural origin of a scenario is expensive, and the honest reading of the Hormuz tiers is directional. A closure of the strait would be clearly damaging for Vietnam, contractionary but imprecisely so for the Philippines and China, and close to neutral or mildly positive for the rest of the region, while a demand-driven rise of comparable size would be expansionary for most of it. The comparison of Figures 10 and 11 is the paper's central lesson stated with two concrete events. What matters for the region is not that the oil price moves but the shock that moves it.

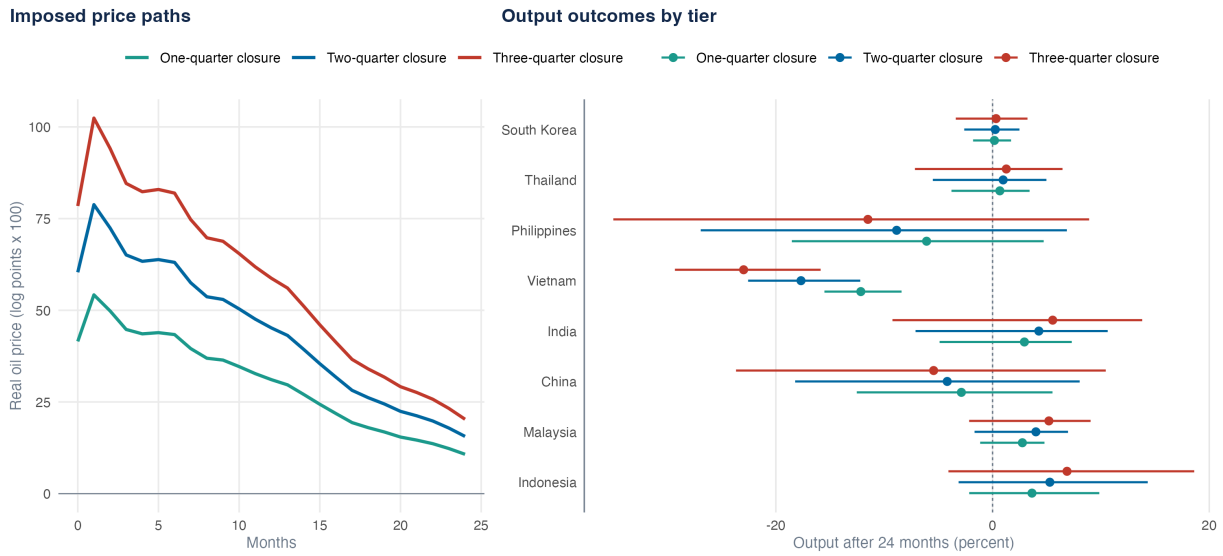


Figure 11: A closure of the Strait of Hormuz, three tiers by duration of the closure

Notes. The left panel shows the imposed real oil price paths, each a single flow-supply shock scaled so that its peak matches the corresponding closure tier of Kilian, Plante, Richter, and Zhou (2026). The right panel shows the cumulative output response after 24 months for each economy and tier, median and 68 percent band. Economies are ordered from most energy-import-intensive to most export-intensive. *Source.* Authors' structural VAR.

The eight economies also face country-specific risks that differ in their driver and in their size. We design one tailored scenario for each economy, ranging from a fuel-import cost shock for the Philippines to a commodity export windfall for Indonesia, and we report the full set in Appendix B. The complete set of block-identified impulse responses for all eight economies is in Appendix C.

5 Conclusion

This paper has asked how oil shocks move output and prices in eight Asian economies, and it has answered the question shock by shock and country by country. The main lesson is that the source of an oil price change is what matters. A demand-driven increase, which arrives with a stronger world economy, raises output in most of the region, while a pure supply-driven increase is smaller and more heterogeneous and does not line up neatly with each economy's position in the energy trade. Oil is a first-order driver of the regional business cycle throughout, and a concrete disruption such as a closure of the Strait of Hormuz would fall unevenly across the region, with the sharpest contractions in Vietnam and the Philippines. Two extensions would sharpen the analysis. The first is to design intermediate oil price paths whose plausibility is closer to the range of ordinary experience than the stylized scenarios

used here. The second is to bring in the role of administered fuel prices highlighted by Cuñado, Jo, and Pérez de Gracia (2015), which is likely to matter for Indonesia and for the interpretation of the inflation channel across the region.

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Data note. The data come from the Federal Reserve Economic Data service at https://fred.stlouisfed.org/, the Index of Global Real Economic Activity of the Federal Reserve Bank of Dallas at https://www.dallasfed.org/research/igrea, the United States Energy Information Administration at https://www.eia.gov/, the International Financial Statistics of the International Monetary Fund at https://data.imf.org/, the World Bank World Development Indicators at https://data.worldbank.org/, and the Philippine Statistics Authority at https://openstat.psa.gov.ph/.

A Descriptive statistics

Table 5 reports descriptive statistics for each economy over its estimation sample. The variables and their sources are described in Section 3.

Table 5: Descriptive statistics by economy

Economy	Sample	Obs.	Ind. prod. growth	Inflation (year on year)	Exch. rate change	Policy rate
Philippines	2001m12–2024m04	269	2.1 (35.2)	3.8 (2.0)	0.4 (6.4)	4.8
Malaysia	2001m12–2024m04	269	3.5 (6.4)	2.1 (1.7)	1.0 (6.8)	3.1
Thailand	2001m01–2024m04	280	2.9 (9.5)	2.0 (2.5)	-1.0 (5.9)	2.0
Indonesia	2001m01–2024m03	236	3.5 (6.0)	6.5 (8.5)	2.2 (8.7)	7.0
Vietnam	2008m01–2023m12	188	0.5 (19.5)	5.2 (4.4)	2.6 (3.1)	7.3
India	2006m05–2022m07	195	3.9 (11.2)	6.7 (2.7)	3.4 (8.1)	7.5
China	2001m01–2024m04	280	9.8 (5.3)	2.2 (1.9)	-0.6 (4.4)	5.3
South Korea	2001m01–2024m04	280	3.8 (6.5)	2.4 (1.3)	0.1 (9.7)	2.8

Notes. Industrial production growth, inflation, and the exchange-rate change are year-on-year percent changes, reported as the sample mean with the standard deviation in parentheses. The exchange rate is local currency per United States dollar, so a positive change is a depreciation. The policy rate is the sample mean of the level in percent. *Source.* Authors' calculations.

B Country-specific scenarios

This appendix reports one tailored scenario for each economy. Each scenario is a single structural oil shock that raises the real price of oil by twenty percent at its peak, with the driver chosen to match the exposure that is most relevant for the economy in question, and each is graded with the Kullback and Leibler plausibility statistic q computed for the decaying price path that the figures display. Table 6 collects the designs and the two-year outcomes, and Figures 12 through 19 report the trajectories of the four domestic variables. The paragraphs below discuss each case in turn.

South Korea. South Korea imports more than four fifths of the energy it uses and hosts one of the world's largest refining and petrochemical complexes, so an oil-supply disruption is the natural risk to examine. The estimated consequences are nonetheless small, with output at +0.1 percent and consumer prices at -0.3 percent after two years, and neither band excludes zero. The muted response of an economy this dependent on imported energy suggests considerable capacity to absorb input-cost shocks, and it cautions against equating import dependence with macroeconomic vulnerability.

Thailand. Thailand combines a high energy import bill with an economy in which tourism and trade are large, both of which are sensitive to fuel costs. Under the supply-driven scenario output is essentially unchanged at +0.2 percent while consumer prices rise by +0.7 percent, so the burden of the shock appears in prices rather than in activity. Thailand thus sits in the middle of the regional spectrum, hurt too little to register in output yet exposed enough for the cost channel to show.

Philippines. The Philippines produces almost no crude oil and imports nearly all of its petroleum requirement, which makes the supply-driven scenario a direct test of the import-cost channel. Output falls by -2.2 percent after two years, among the clearest contractions in the region, while consumer prices are little changed at -0.3 percent. The output response rather than the price response carries the burden, consistent with an economy in which fuel is an intermediate input for transport and power more than a driver of the consumption basket.

Vietnam. Vietnam is the region's fastest-growing manufacturing exporter, and its production is intensive in energy and in imported intermediate inputs. The supply-driven scenario is accordingly punishing, with output lower by -4.5 percent, the largest contraction among the eight economies, and consumer prices higher by +1.2 percent. Vietnam is the one economy in which the scenario is stagflationary in the textbook sense, with activity falling while prices rise.

India. India imports the great majority of its crude oil and its currency has historically come under pressure in oil price spikes, so we examine the supply-driven scenario. The estimated output response is positive at +1.1 percent but imprecise, and prices are little changed at -0.4 percent. The absence of the expected contraction echoes the finding of Cuñado, Jo, and Pérez de Gracia (2015) that administered fuel prices and subsidies blunt the pass-through of oil shocks in India, and it counsels against imposing the importer prior on the data.

China. For China the relevant experiment is not an external cost shock but a demand cycle of its own making, since Chinese demand is itself a principal driver of the world oil price, as Aastveit, Bjørnland, and Thorsrud (2015) document. Under the joint model, in which feedback from the Chinese economy to the oil market is permitted, a global-demand shock that raises the oil price by twenty percent leaves Chinese output at +0.2 percent and prices at +0.6 percent. The near-neutral outcome is consistent with a shock that is partly

China's own boom returned to it as a higher input cost.

Malaysia. Malaysia is close to energy balance, exporting liquefied natural gas and refined products while importing crude, and its export basket is commodity-intensive well beyond energy. The demand-driven scenario delivers a terms-of-trade gain, with output higher by +0.7 percent and consumer prices higher by +1.0 percent. The gain is modest because the higher energy cost offsets part of the commodity windfall.

Indonesia. Indonesia is the region's clearest commodity exporter, with coal, gas, palm oil, and metals dominating its export earnings, and it left the Organization of the Petroleum Exporting Countries as its own crude production declined. The demand-driven scenario produces the largest gain in the region, with output higher by +2.3 percent and consumer prices higher by +2.4 percent, and it is also the most plausible scenario in the set, with q equal to 0.66. A global boom that raises the oil price raises the price of nearly everything Indonesia sells, and the economy expands accordingly even as inflation picks up.

Table 6: Country-specific scenarios, design and two-year outcomes

Economy	Scenario	Output	Prices	q
South Korea	Energy-import cost shock	+0.1	-0.3	0.89
Thailand	Fuel and tourism cost shock	+0.2	+0.7	0.89
Philippines	Fuel-import cost shock	-2.2	-0.3	0.87
Vietnam	Manufacturing input-cost shock	-4.5	+1.2	0.83
India	Oil-import and currency shock	+1.1	-0.4	0.92
China	China-led global demand cycle	+0.2	+0.6	0.79
Malaysia	Commodity terms-of-trade gain	+0.7	+1.0	0.76
Indonesia	Commodity export windfall	+2.3	+2.4	0.66

Notes. Output and prices are the cumulative percent response after 24 months. Each scenario is a single shock that raises the real oil price by twenty percent at its peak, of the driver type indicated by the scenario. China uses the joint model of Section 4. *Source.* Authors' structural VAR.

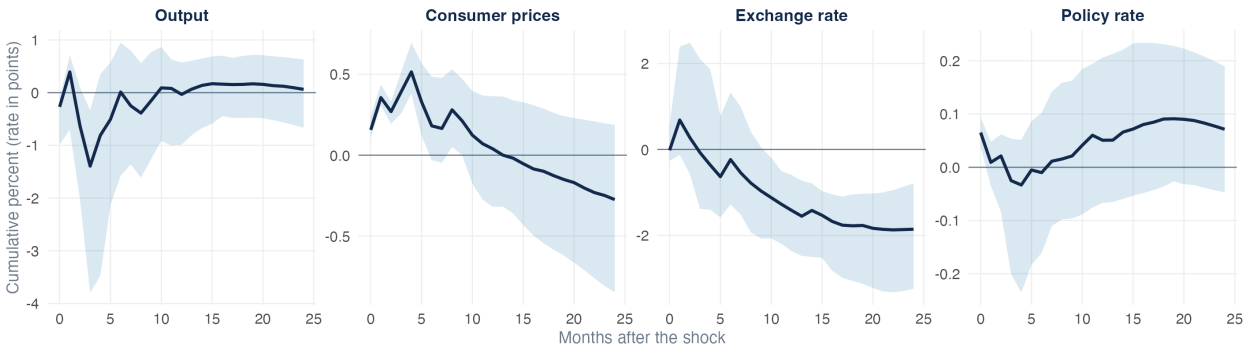


Figure 12: South Korea, energy-import cost shock

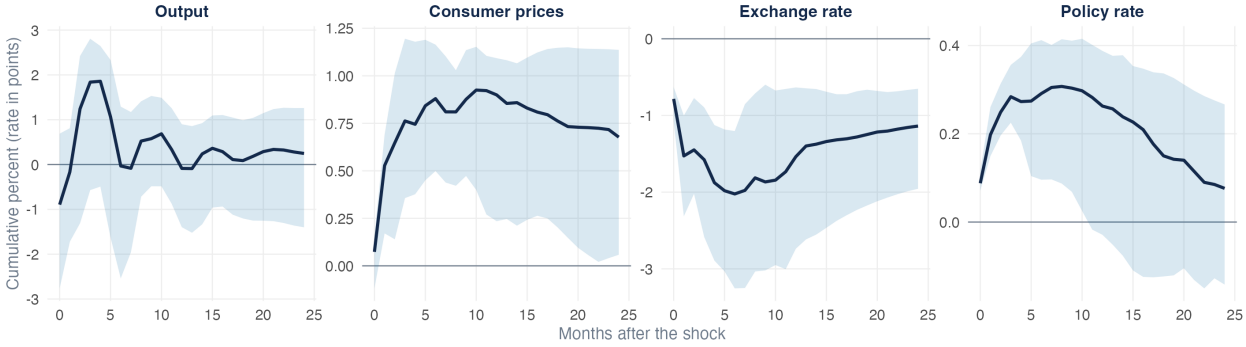


Figure 13: Thailand, fuel and tourism cost shock

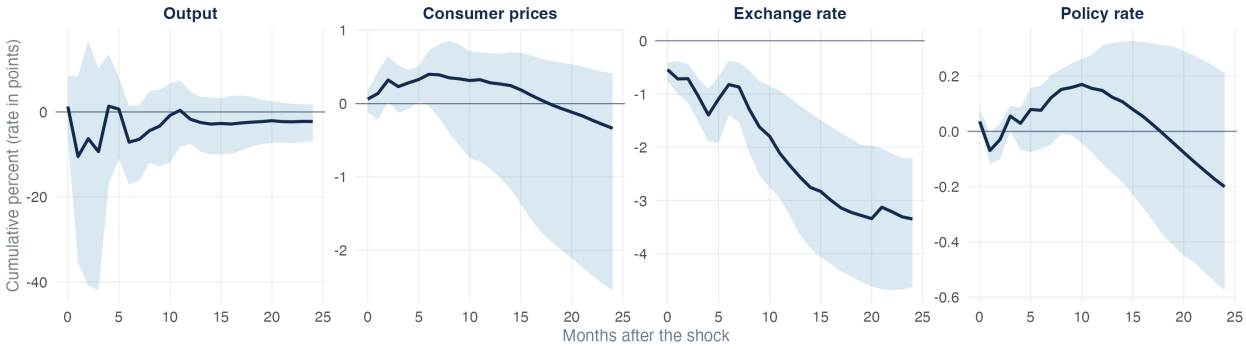


Figure 14: Philippines, fuel-import cost shock

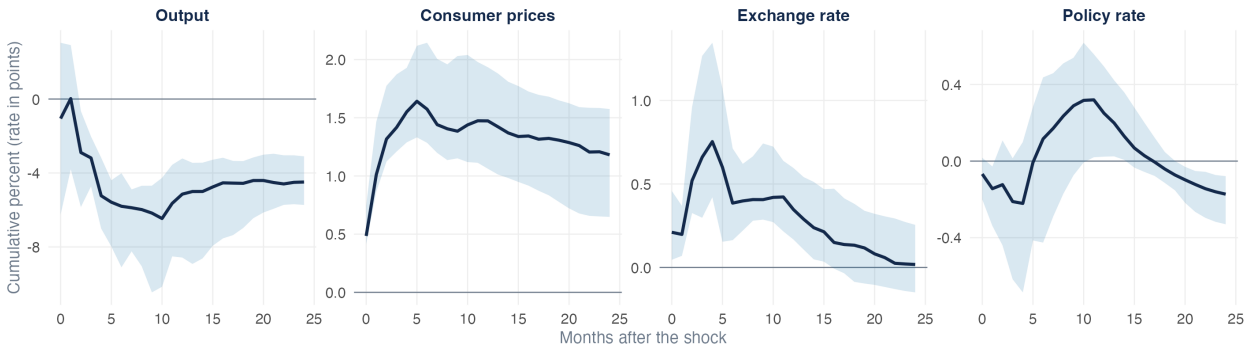


Figure 15: Vietnam, manufacturing input-cost shock

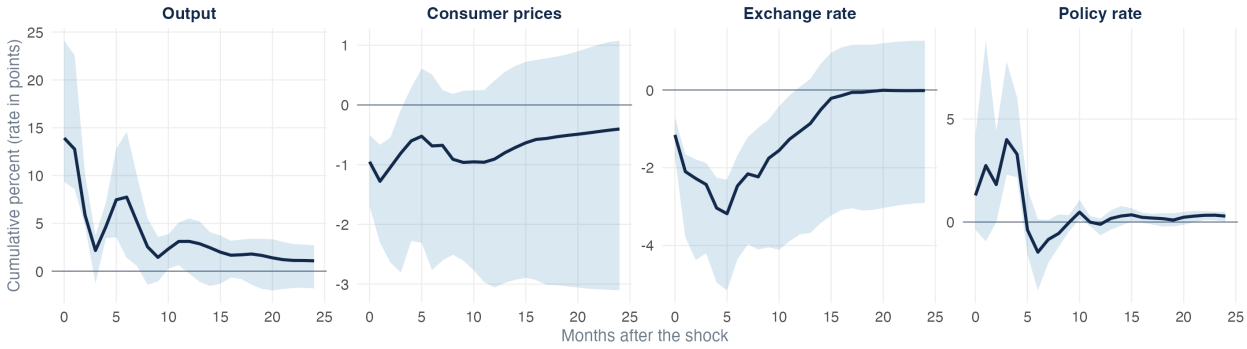


Figure 16: India, oil-import and currency shock

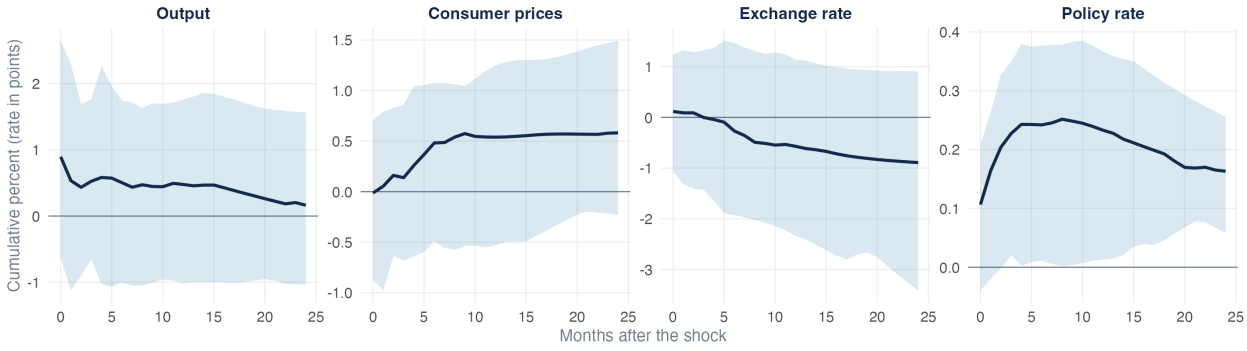


Figure 17: China, global-demand cycle, joint model

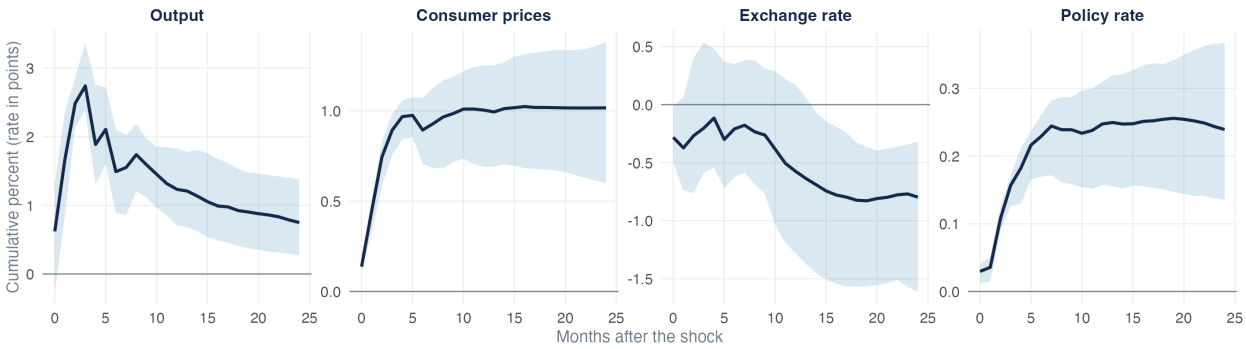


Figure 18: Malaysia, commodity terms-of-trade gain

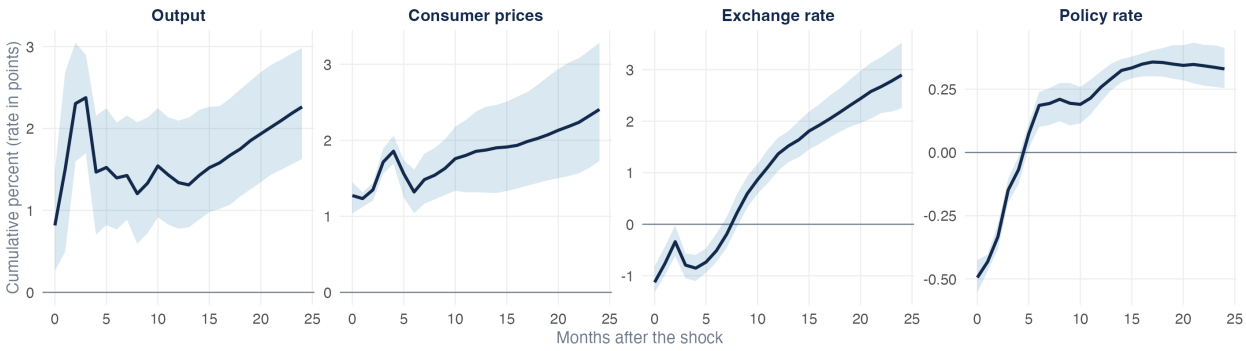


Figure 19: Indonesia, commodity export windfall

C Impulse responses for all economies

This appendix reports the block-identified impulse responses of the four domestic variables to the three oil shocks for each economy, over 24 months, with 68 percent credible bands. The responses use the block-exogeneity identification for every economy except China, which uses the joint model. Each figure places the domestic variables in rows and the oil shocks in columns.

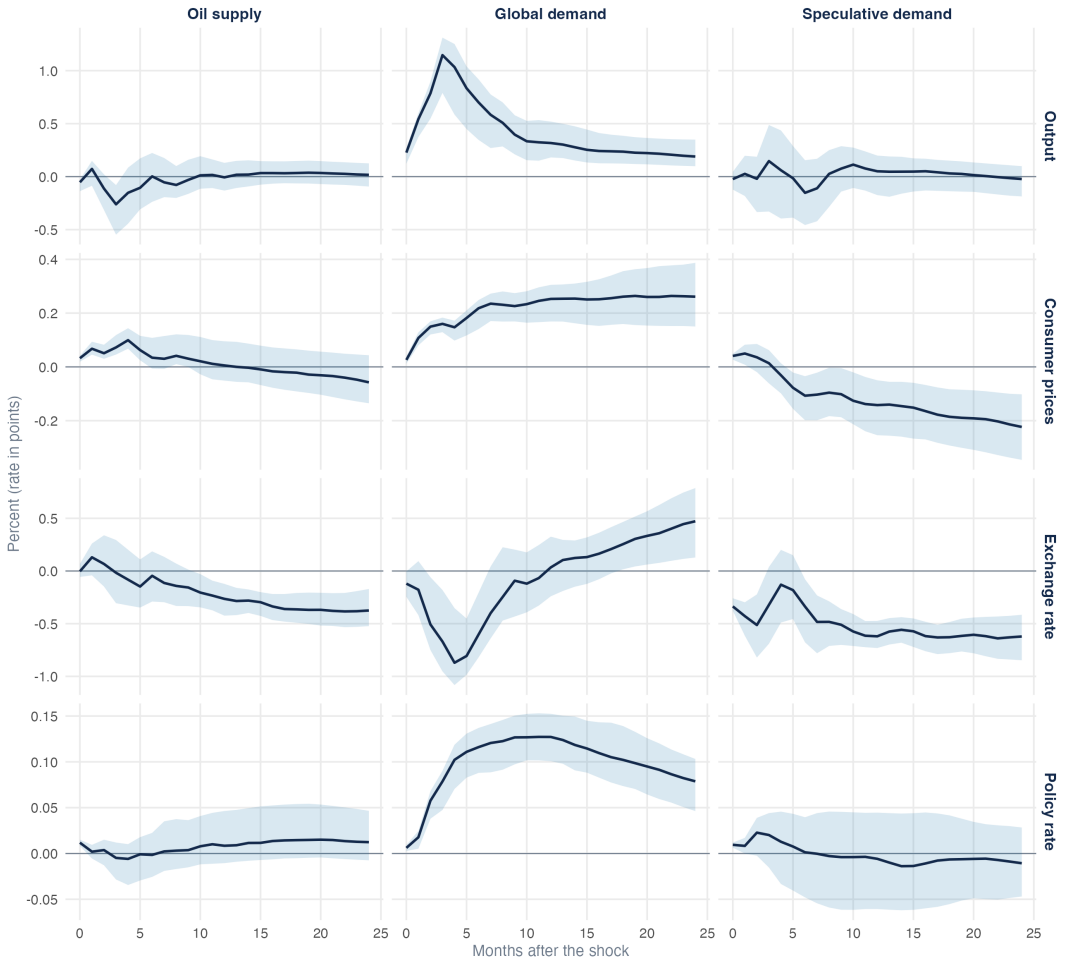


Figure 20: South Korea, responses to the three oil shocks

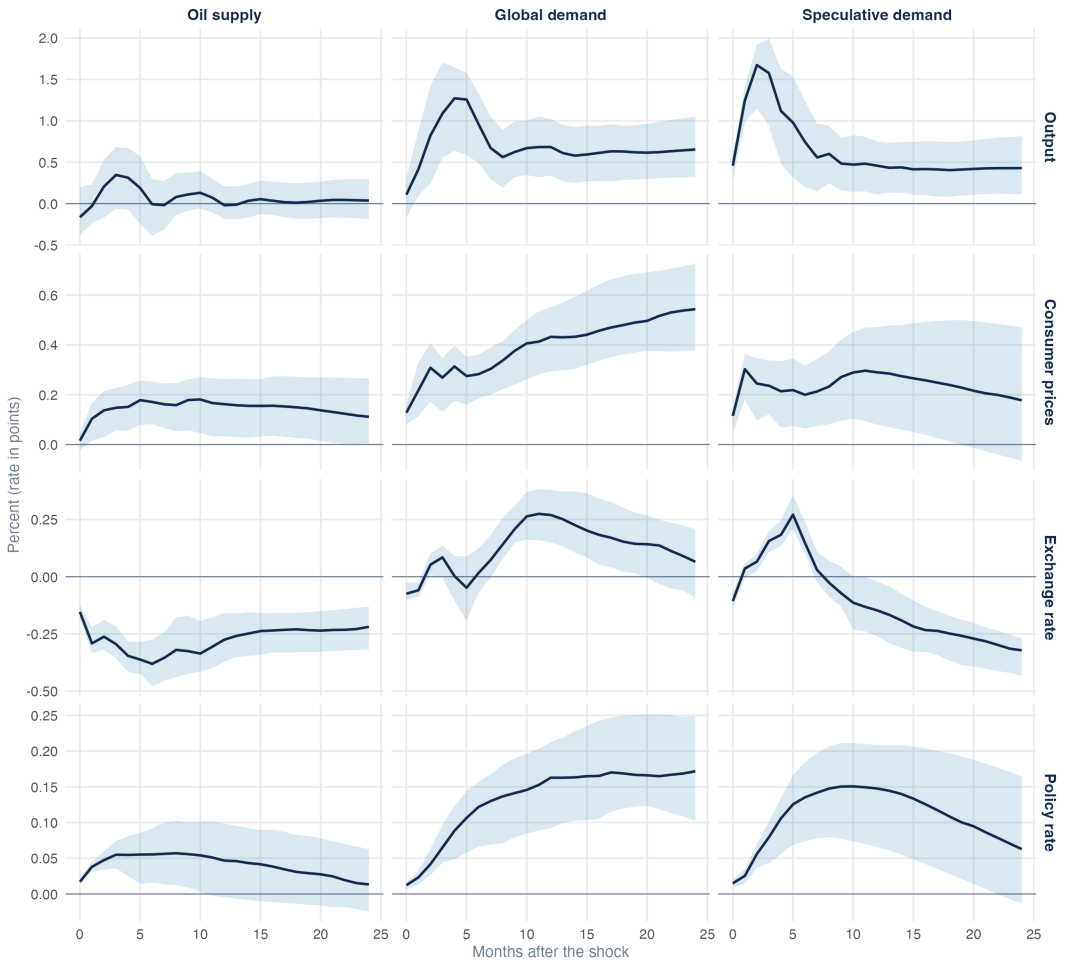


Figure 21: Thailand, responses to the three oil shocks

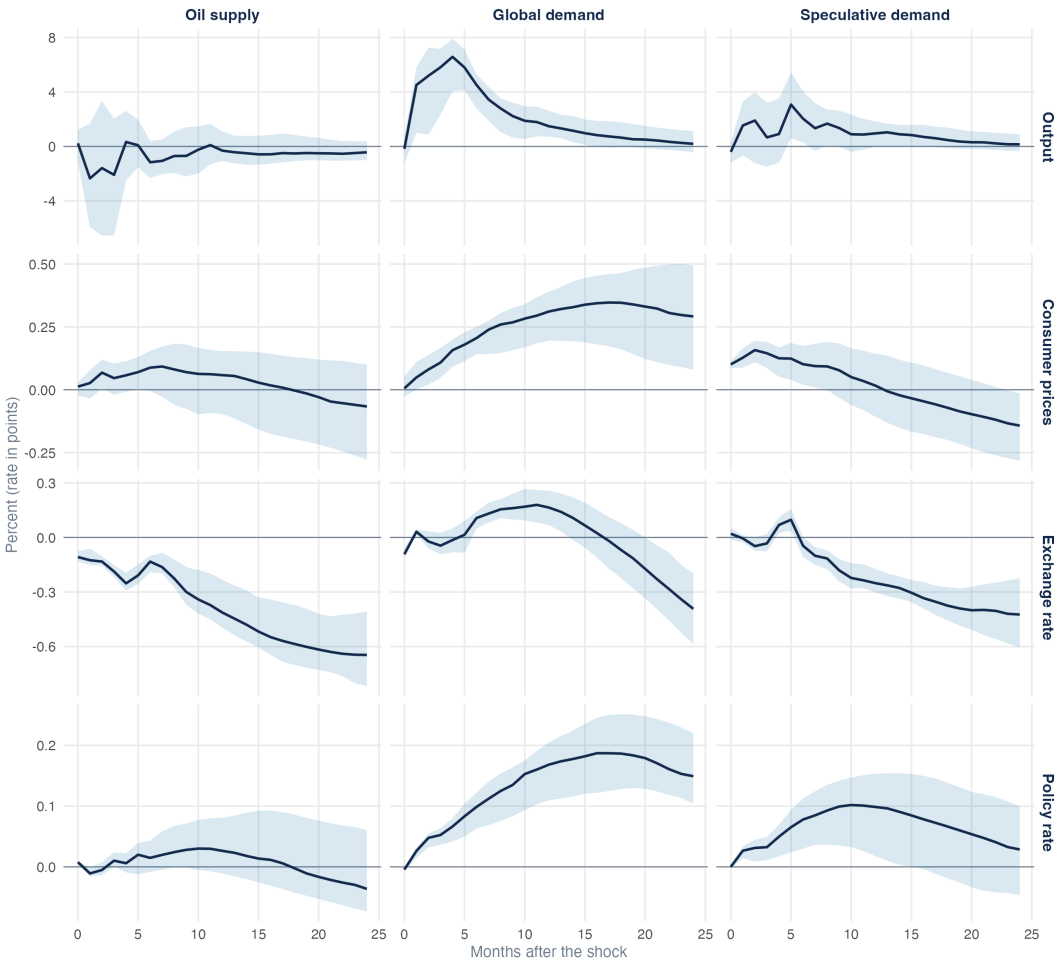


Figure 22: Philippines, responses to the three oil shocks

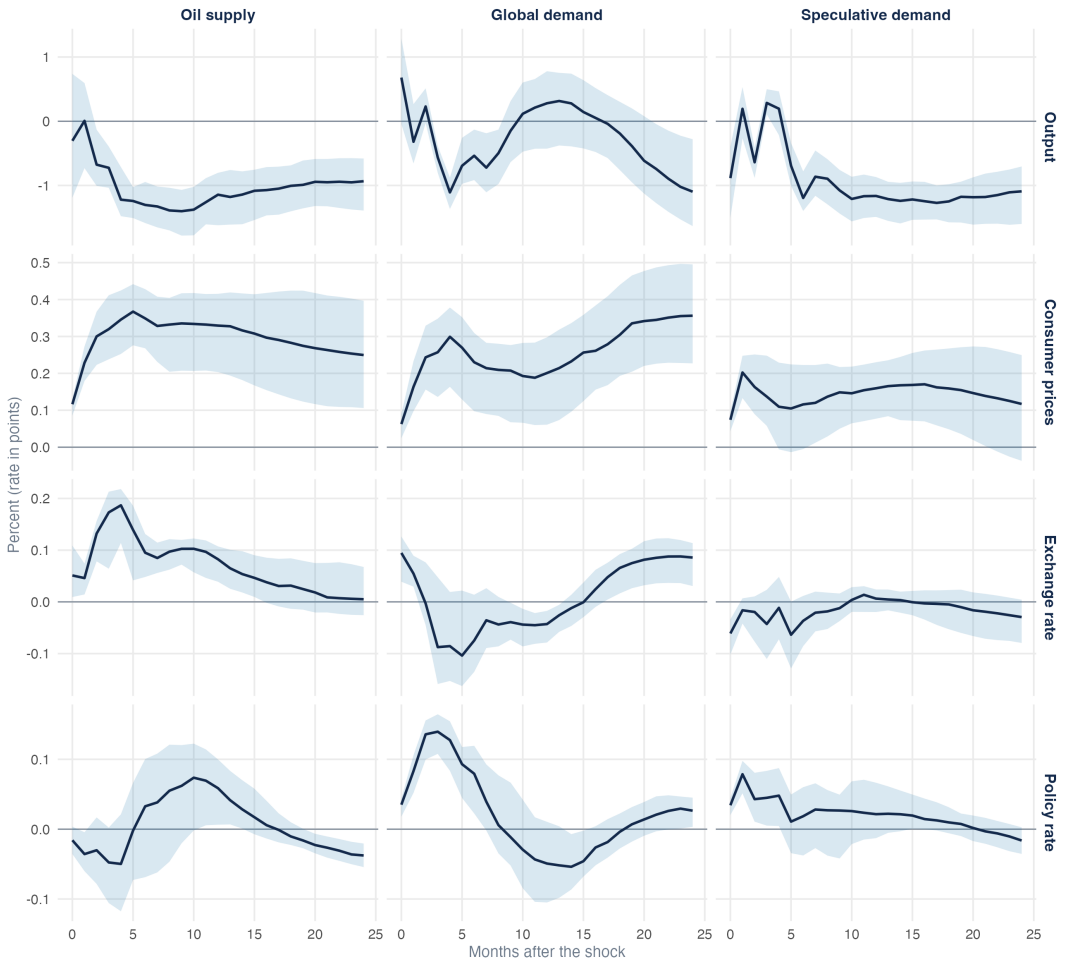


Figure 23: Vietnam, responses to the three oil shocks

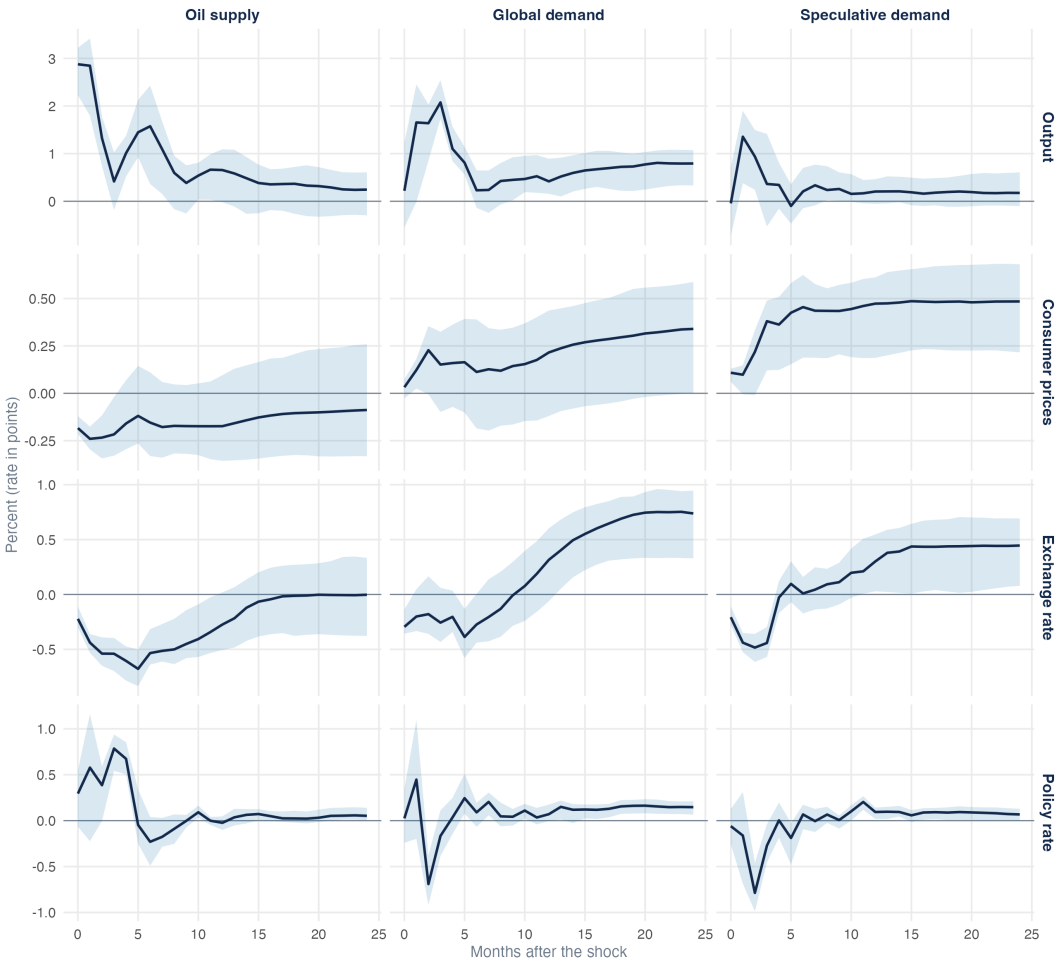


Figure 24: India, responses to the three oil shocks

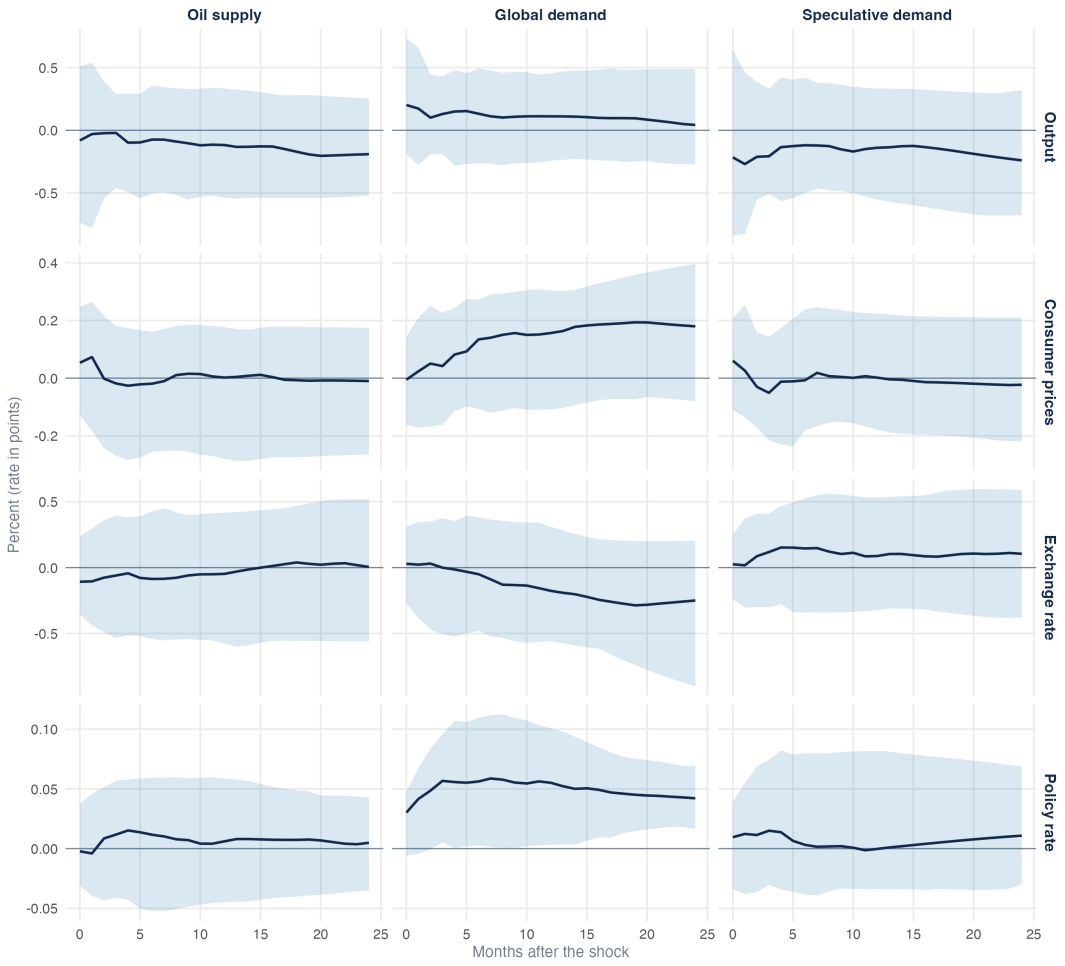


Figure 25: China, responses to the three oil shocks, joint model

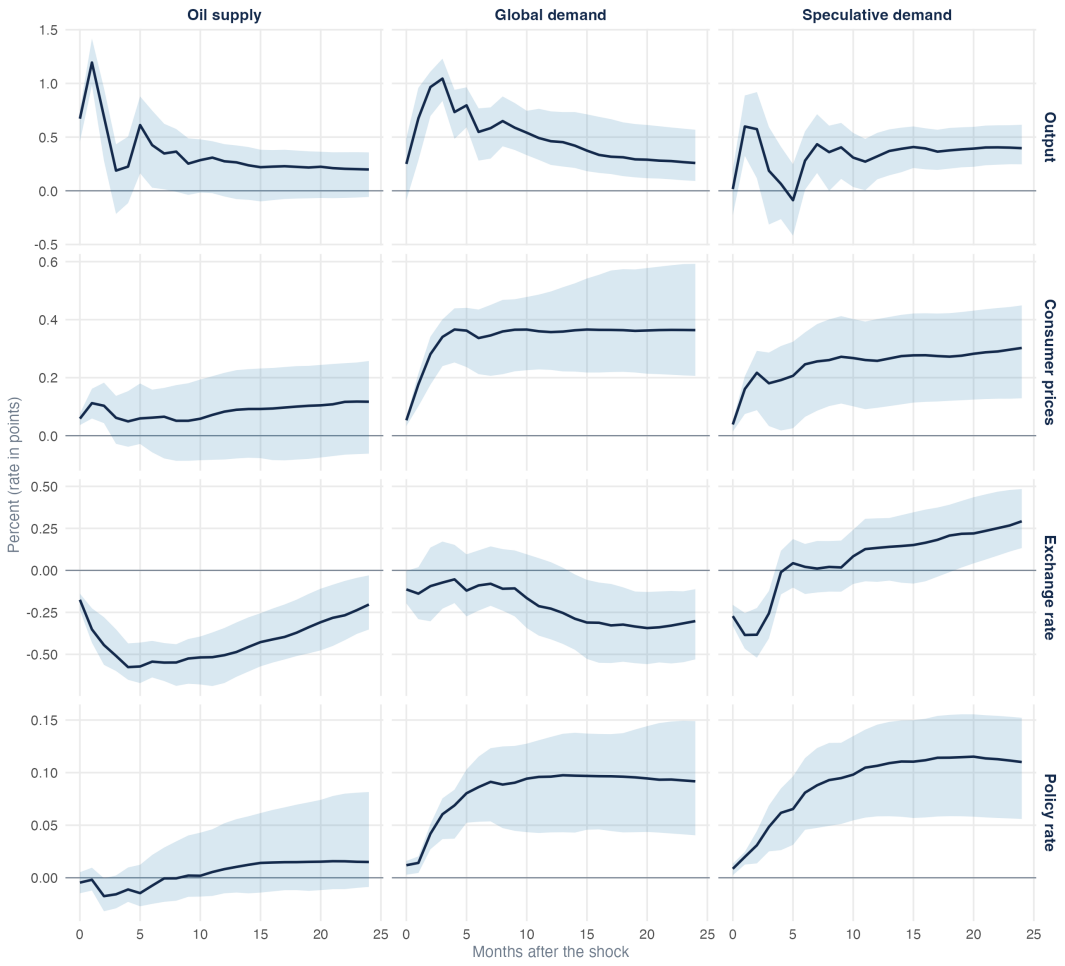


Figure 26: Malaysia, responses to the three oil shocks

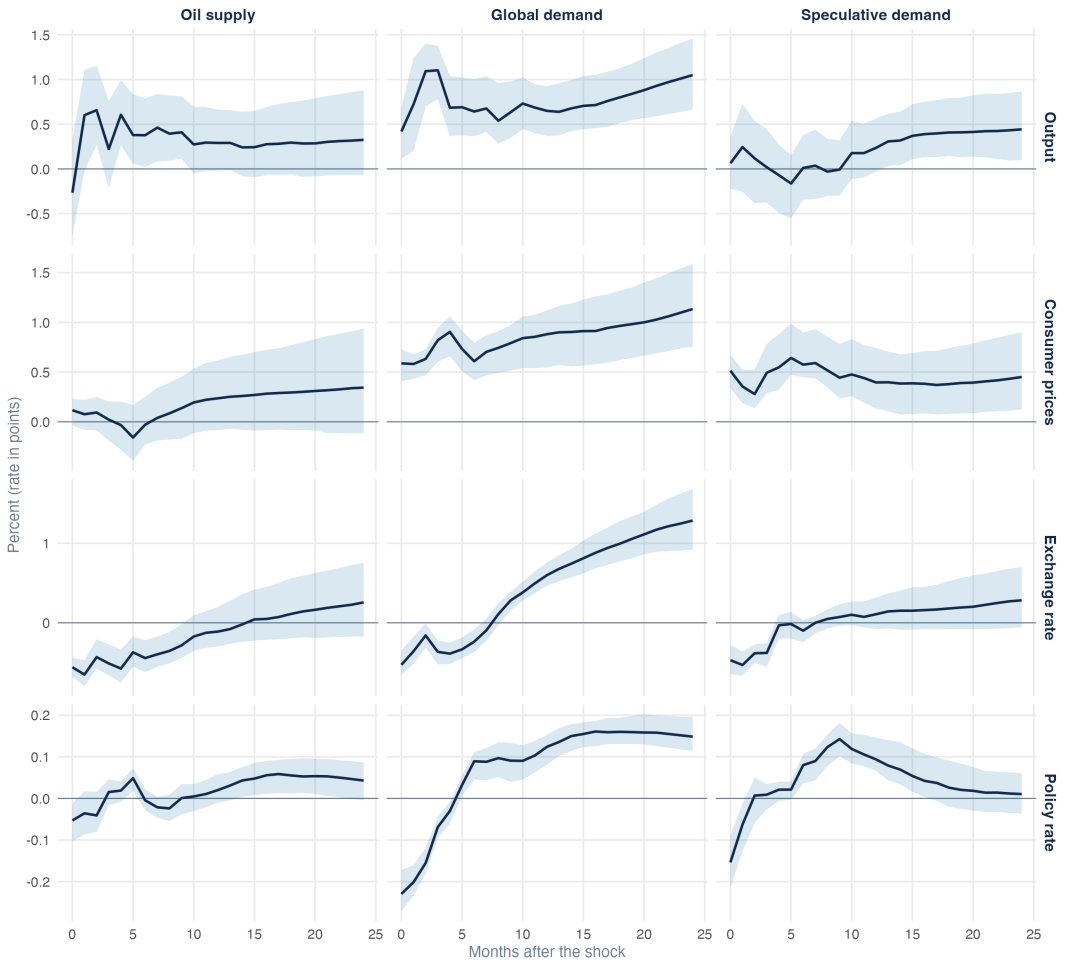


Figure 27: Indonesia, responses to the three oil shocks